Chapter 7

Derivation of Sines

7.1 The side of a regular hexagon inscribed in a circle is equal to the radius

It is instructed that using the methods of the previous chapter the diameter of a circle whose circumference is given by 21,600 minutes be calculated; and halving the diameter the corresponding radius be found.[1] Then we are instructed to draw a circle of this radius.

In Figure 7.1 we consider a circle and draw the east-west line EW and north-south line NS through its centre O. Draw the full-chords NA, ND from the north point and SB, SC from the south point of length equal to the radius. Join the radii OA, OD, OB and OC. We then have four equilateral triangles OAN, ODN, OBS and OCS with sides equal to the radius. From A, B, drop the perpendiculars (*lamba*) AP, BQ to the north-south line. Since these bisect the base, it follows that

$$NP = PO = OQ = QS = \frac{r}{2}\ . \tag{7.1}$$

Hence, it follows that

$$AB = PQ = r. \tag{7.2}$$

Similarly it follows that $CD = r$. Hence it follows that $NABSCD$ is a regular hexagon, of side equal to the radius, which is inscribed in the circle.

[1]The standard units employed are 1 *bhāga* (degree) = 60 *kalā* (*liptā* or *ili*, minute); 1 *kalā* = 60 *vikalā* (second) and so on. If the diameter is $21600'$, then a fairly accurate value of radius is $3437'44''48'''22^{iv}29^{v}22^{vi}22^{vii}$ given in the *katapayādi* notation by the verse *śrīrudraḥ śridharaḥ śreṣṭho devo viśvasthalī bhṛguḥ*.

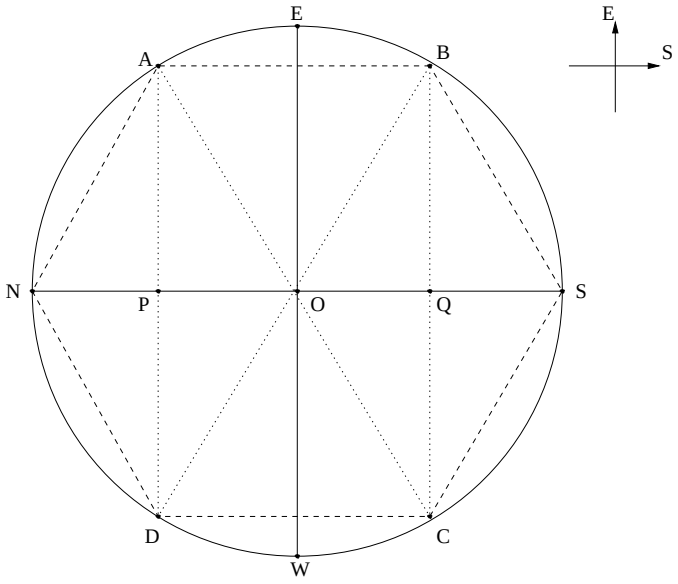


Figure 7.1: Regular hexagon inscribed in a circle.

Hence the *samasta-jyā* or the full-chord,[2] of two *rāśi*-s (60°) is the radius and hence Rsine (*ardha-jyā* or just *jyā*) of one *rāśi* is given by

$$Jyā(EA) = \frac{r}{2} \,. \tag{7.3}$$

7.2 Derivation of Rsines

7.2.1 *Jyā, Koṭi* and *Śara* : Rsine, Rcosine and Rversine

Half the full-chord (*samasta-jyā*) of an arc is the half-chord (*ardha-jyā*) of half the arc. Since in planetary astronomy it is only the *ardha-jyā* that is used, it will be called just *jyā*, or Rsine of the half-arc.

In Figure 7.2, the arc EA is one *rāśi* (sign, arc corresponding to 30°). Then AP, the half-chord of double the arc is the *jyā* or Rsine. The Rsine of the

[2]The word *jyā* refers to the 'string' of a bow. In Figure 7.1 if AEB is considered as a bow, then the chord AB can be conceived of as the string of the bow.

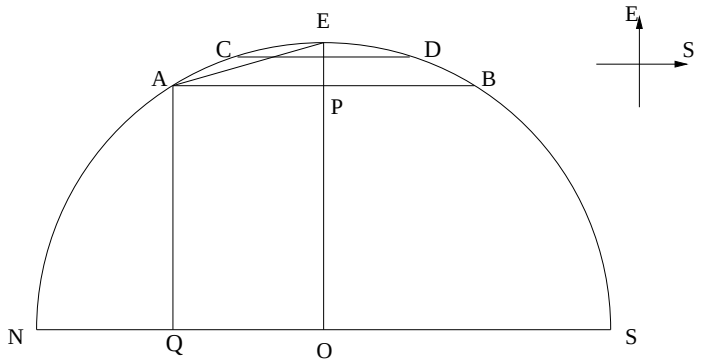


Figure 7.2: *Jyā*, *Koṭi* and *Śara*.

arc NA (the complimentary arc of EA), namely AQ, is the *koṭi* or Rcosine. If we mark the Rcosine along the east-west line as PO, then the bit of the radius remaining, PE, is the *śara* or Rversine.[3] If we mark the Rsine along the north-south line as OQ, then the bit of the radius remaining, QN, is the *koṭi-śara*, or the Rvercosine.

If $s = r\theta$, is the arc of a circle of radius r (in minutes), subtending angle θ (in radians) at the centre, then

$$\begin{aligned} \textit{Jyā}(s) &\equiv R\sin(s) = r\sin(\theta) = r\sin\left(\frac{s}{r}\right), && (7.4)\\ \textit{Koṭi-jyā}(s) &\equiv R\cos(s) = r\cos(\theta) = r\cos\left(\frac{s}{r}\right), && (7.5)\\ \textit{Śara}(s) &\equiv R\ \mathrm{vers}(s) = r\ \mathrm{vers}(\theta) = r\ \mathrm{vers}\left(\frac{s}{r}\right)\\ &= r\left(1-\cos\left(\frac{s}{r}\right)\right). && (7.6) \end{aligned}$$

In most of what follows, it is assumed that the circumference of the circle is $21,600'$ or $360°$.

[3]The term *śara* means 'arrow'. In Figure 7.2, if we consider the arc AEB as a bow, then PE serves as *śara* or arrow joining the mid point of the full-chord with the middle of the arc AB.

7.2.2 Derivation of Rsines

Now, in Figure 7.2, consider the triangle APE, where AP, the Rsine, is the *bhujā* and PE, the Rversine, is the *koṭi*. Hence the full-chord (*samasta-jyā*) of one sign, AE, is given by

$$AE^2 = AP^2 + PE^2. \tag{7.7}$$

But the full-chord AE of one sign is twice the Rsine of half a sign. Thus

$$R\sin(EC) = \left(\frac{1}{2}\right)\left[R\sin^2(EA) + R\ \mathrm{vers}^2(EA)\right]^{\frac{1}{2}}. \tag{7.8}$$

The same relation will hold of any arc s:

$$R\sin\left(\frac{s}{2}\right) = \left(\frac{1}{2}\right)\left[R\sin^2(s) + R\ \mathrm{vers}^2(s)\right]^{\frac{1}{2}}. \tag{7.9}$$

There is also the standard relation

$$R\cos(s) = R\sin(90^\circ - s) = \left[r^2 - R\sin^2(s)\right]^{\frac{1}{2}}, \tag{7.10}$$

where we have assumed that 90° is one-fourth the circumference. Equations (7.9) and (7.10) can be used along with the standard values[4]

$$R\sin(30^\circ) = \frac{r}{2}, \tag{7.11}$$

$$R\sin(45^\circ) = \left(\frac{r^2}{2}\right)^{\frac{1}{2}}, \tag{7.12}$$

to derive a number of Rsines.[5]

[4]The $R\sin 30^\circ$ value has been derived in (7.3). The value of $R\sin(45^\circ)$ can be obtained by noting that the square of the full-chord (EN in Figure 7.2) of the corresponding double-arc of 90°, is given by $2r^2$.

[5]As will be explained later in Section 7.4.2, the object is to derive the 24 tabulated Rsines of arcs which are multiples of $225' = 3^\circ 45'$. Here the starting value (7.11) of $R\sin(30^\circ)$ is the 8th tabulated sign. From this, the Rsine of the complementary arc 60°, or the 16th tabulated Rsine, is obtained from (7.10). The Rsine of the half-arc, $R\sin 15^\circ$, or the 4th tabulated Rsine is found using (7.9). In this way, by successive applications of (7.9) and (7.10), the 1st, 2nd, 4th, 5th, 7th, 8th, 10th, 11th, 13th, 14th, 16th, 17th, 19th, 20th, 22nd, 23rd and 24th tabulated Rsines are all determined. The remaining tabulated Rsines are found by following the same procedure, starting from the value (7.12) of $R\sin(45^\circ)$, which is the 12th tabulated Rsine.

7.3 Some technical terms and definitions

7.3.1 Rsine and Rcosine

Consider the quadrant of the circle from the east point to the north point. This quadrant may be divided into a convenient number, usually taken to be 24, of equal *cāpa-khaṇḍa*-s or arc-bits.

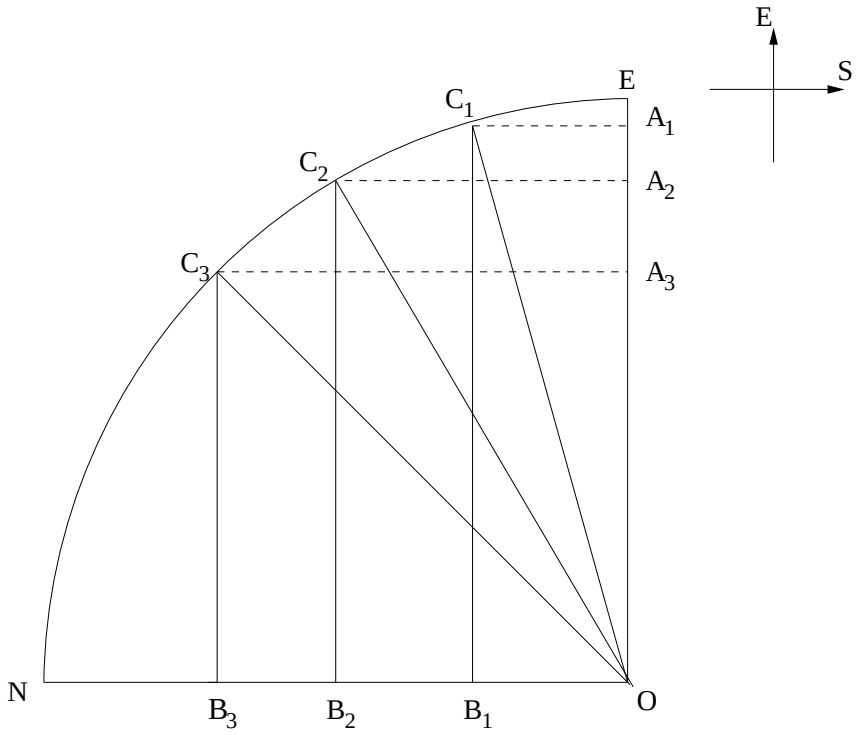


Figure 7.3: Rsines and Rcosines.

In Figure 7.3 are shown (not to scale) the first three arc-bits, EC_1, C_1C_2 and C_2C_3. The Rsines (*bhujā-jyā*) are parallel to the north-south line and the Rcosines (*koṭi-jyā*) and the Rversines (*śara* or *utkrama-jyā*) are parallel to the east-west line. C_1A_1, C_2A_2, C_3A_3 are the Rsines of the arcs EC_1, EC_2, and EC_3 respectively. C_1B_1, C_2B_2, C_3B_3 are the Rcosines of these arcs, and A_1E, A_2E, and A_3E are the Rversines of these arcs, respectively.

7.3.3 Rsine and Rcosine-differences

The Rsine-difference (*bhujā-khaṇḍa*) of the first arc-bit EC_1 is B_1O, which is the same as C_1A_1, the first Rsine itself. The Rsine-difference of the second arc-bit C_1C_2 is B_2B_1, which is equal to the portion of the second Rsine enclosed between its tip C_2 and the first Rcosine C_1B_1. The Rsine-difference of the third arc-bit C_2C_3 is B_3B_2, which is equal to the portion of the third Rsine enclosed between its tip C_3 and the second Rcosine C_2B_2.

The Rcosine-difference (*koṭi-khaṇḍa*) of the first arc-bit EC_1 is EA_1, which is the distance from the east point to the foot of the first Rsine. The Rcosine-difference of the second arc-bit C_1C_2 is A_1A_2, which is equal to the portion of the first Rcosine enclosed between its tip C_1 and the second Rsine C_2A_2. The Rcosine-difference of the third arc-bit C_2C_3 is A_2A_3, which is equal to the portion of the second Rcosine enclosed between its tip C_2 and the third Rsine C_3A_3. It is noted later that EA_1 is the first Rversine-difference (*śara-khaṇḍa*), which is also the first Rversine. A_1A_2 and A_2A_3 are the next two Rversine-differences.

Each of the Rsine-difference and Rcosine-difference pairs, B_1O and EA_1, B_2B_1 and A_1A_2, B_3B_2 and A_2A_3, will be *bhujā*-s and *koṭi*-s, with the corresponding full-chords of the arc-bit as the *karṇa*. Since the arc-bits are all equal, the corresponding full-chords (*samasta-jyā*-s) are all equal, though in each case, the *bhujā*-s and *koṭi*-s, the Rsine- and Rcosine-differences are all different. This is quite like the Rsines and Rcosines themselves which can be considered as *bhujā*-s and *koṭi*-s which, though differ from arc to arc, are such that the corresponding *karṇa* (the hypotenuse, which is the root of sum of their squares) is always equal to the radius.

7.3.4 Rsine and Rcosine in the quadrants

The east point on the circle is usually taken as the *Meṣādi*, the First Point of Aries, and from there the *rāśi*-s, the signs, are marked towards the north, such that the end of the *rāśi Mithuna* (Gemini) is at the north point. When a given arc (*iṣṭa-cāpa*) is reckoned from the east point and its tip is in different quadrants, then, the correspoinding *bhujā-cāpa* and *koṭi-cāpa* are given as follows (see Figure 7.4).

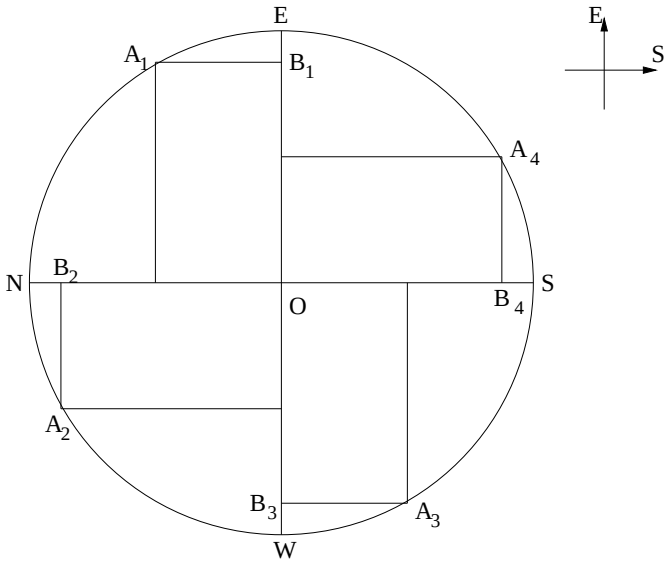


Figure 7.4: *Bhujā-cāpa* and *Koṭi-cāpa* in different quadrants.

In the first quadrant, if the given arc is EA_1, then EA_1 itself is the *bhujā-cāpa* and NA_1 is the *koṭi-cāpa*. In the second quadrant, if the given arc is EA_2, then WA_2 is the *bhujā-cāpa* and A_2N is the *koṭi-cāpa*. In the third quadrant, if the given arc is EA_3, then WA_3 is the *bhujā-cāpa* and A_3S is the *koṭi-cāpa*. In the fourth quadrant, if the given arc is EA_4, then EA_4 is the *bhujā-cāpa* and A_4S is the *koṭi-cāpa*.

7.4 Computation of Rsines

7.4.1 Tabular Rsines (*Paṭhita-jyā*)

Having divided the quadrant into a number of equal arc-bits, the corresponding Rsine-differences may be calculated and tabulated. These are called tabulated Rsine (differences), *paṭhita-jyā*, as has been done in earlier texts (*pūrva-śāstra*-s).[6] They can be tabulated in order, or in the inverse order as *utkrama-jyā*-s.

[6]Some texts such as the *Āryabhaṭīya* tabulate Rsine-differences, while other texts such as the *Sūryasiddhānta* tabulate the Rsines themselves.

If a given arc is a multiple of these arc-bits, then the corresponding Rsine can be found merely by adding these tabulated Rsine-differences. But if the desired arc falls in the interstices of these arc-bits, then a suitable part of the next Rsine-difference is to be added to the value of the Rsine of the nearest arc which is an integral multiple of the arc-bits.

It is noted that we will not get accurate results by just applying the "rule of three" that if an arc-bit corresponds to so much of Rsine-difference, how much a portion of it will correspond to. Such a result will only be rough. This can be seen by noting that while the arc-bits are all equal, the Rsine-differences are all different, so that the Rsine of two arc-bits is not twice the Rsine of a single arc-bit. The reason for this is explained as follows: The first arc is not curved, since the Rversine is very small. So the arc is practically equal to the Rsine. But, as the arc increases the curved nature will increase. There, the Rsine (*bhujā-jyā*) will have lesser length, since the Rversine increases in length. Therefore, the rule of three should not be applied to derive the Rsines.

7.4.2 Computation of accurate tabular Rsines

Before going to the calculation of Rsines when the desired arc falls in the interstice between the arc-bits, a method for calculating the tabulated Rsines is presented. For this purpose, consider the quadrant EN of the circle (of circumference $21600'$) to be divided into 24 equal arc-bits of $225'$ each. We shall show the first two arc-bits in Figure 7.5 where, for the sake of visual clarity, we have taken these arc-bits to be much larger than $225'$. EC_1 and C_1C_2 are the first two arc-bits. Draw the first two Rsines C_1B_1 and C_2B_2, as also the first two Rcosines C_1K_1 and C_2K_2. Further, let C_1K_1 and C_2B_2 intersect at G.

The chords of all the equal arc-bits are all equal and let this chord-length be denoted as α. It is equal to the square-root of the sum of the squares of the first Rsine C_1B_1 and the first Rversine EB_1.

Let M_1, M_2 be the mid-points of the arc-bits EC_1, C_1C_2. Let M_1P_1 and M_1Q_1 be the Rsine and Rcosine of arc EM_1, and M_2P_2 and M_2Q_2, those of arc EM_2. Let H be the point of intersection of M_1Q_1 and M_2P_2.

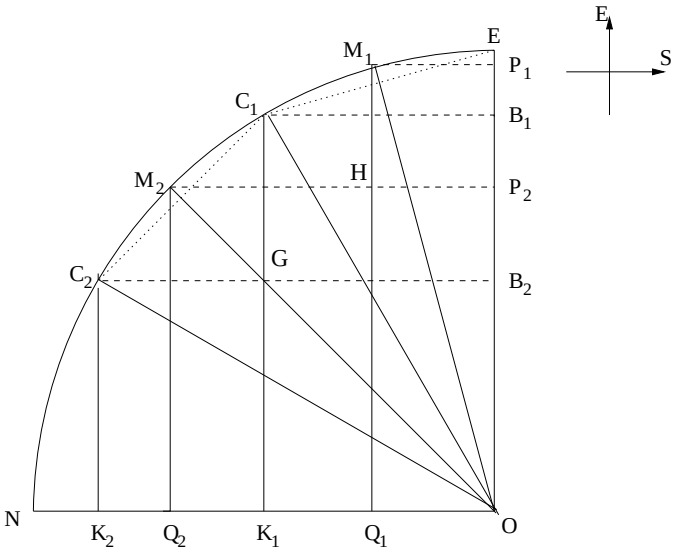


Figure 7.5: Computation of tabular Rsines.

Now, the arc EM_1 is of measure $112\frac{1}{2}'$. Since this arc may be considered small, we can approximate the corresponding Rsine, M_1P_1 by the arc itself. Thus

$$R\sin\left(112\tfrac{1}{2}'\right) = M_1P_1 \approx 112\tfrac{1}{2}'. \tag{7.13}$$

Then, the corresponding Rcosine, M_1Q_1 (which is also the Rsine of $23\frac{1}{2}$ arc-bits of $225'$) and the Rversine, EP_1, are given by

$$\begin{aligned} R\cos\left(112\tfrac{1}{2}'\right) = M_1Q_1 &= \left[r^2 - R\sin^2\left(112\tfrac{1}{2}'\right)\right]^{\frac{1}{2}}, \\ &\approx \left[r^2 - \left(112\tfrac{1}{2}'\right)^2\right]^{\frac{1}{2}}, \qquad (7.14) \\ R\,\mathrm{vers}\left(112\tfrac{1}{2}'\right) = EP_1 &= r - R\cos\left(112\tfrac{1}{2}'\right) \\ &\approx r - \left[r^2 - \left(112\tfrac{1}{2}'\right)^2\right]^{\frac{1}{2}}. \qquad (7.15) \end{aligned}$$

Now, the radius OM_1 bisects the chord EC_1 perpendicularly as M_1 is the midpoint of the first arc-bit. Now the triangles OM_1P_1 and C_1EB_1 are similar as the sides of one are perpendicular to the corresponding sides of the other. Hence, we have

$$\frac{OM_1}{C_1E} = \frac{M_1P_1}{EB_1} = \frac{OP_1}{C_1B_1}\,. \tag{7.16}$$

The chord $C_1E = \alpha$ is twice the Rsine of $112\frac{1}{2}'$. For evaluation of the expressions involving this chord, we shall make the same approximation as in (7.13), that is,

$$\alpha = C_1E = 2\ M_1P_1 \approx 225'. \tag{7.17}$$

From (7.16), we have the following expressions for the first Rsine, Rversine and Rcosine in terms of the values of these quantities for the half-arc-bit $112\frac{1}{2}'$, which have been evaluated above in (7.13)–(7.15):

$$R\sin(225') = C_1B_1 = C_1E \times \frac{OP_1}{OM_1} = \alpha\left[\frac{\left(R\cos\left(112\frac{1}{2}'\right)\right)}{r}\right], \tag{7.18}$$

$$R\ \mathrm{vers}(225') = EB_1 = C_1E \times \frac{M_1P_1}{OM_1} = \alpha\left[\frac{\left(R\sin\left(112\frac{1}{2}'\right)\right)}{r}\right], \tag{7.19}$$

$$R\cos(225') = OB_1 = r - EB_1 = r - \alpha\left[\frac{\left(R\sin\left(112\frac{1}{2}'\right)\right)}{r}\right]. \tag{7.20}$$

Now, in order to calculate the second Rsine, Rversine and Rcosine, we first evaluate the corresponding Rsine-differences etc., between the arcs EM_2 and EM_1. For this purpose we note that the triangles OC_1B_1 and M_2M_1H are similar, Therefore,

$$\frac{OC_1}{M_2M_1} = \frac{OB_1}{M_2H} = \frac{C_1B_1}{M_1H}\,. \tag{7.21}$$

If we note that M_2M_1 is full-chord α, then the Rsine-difference M_2H and Rversine-difference M_1H are given by

$$M_2H = M_2M_1 \times \frac{OB_1}{OC_1} = \alpha\left[\frac{(R\cos(225'))}{r}\right], \tag{7.22}$$

$$M_1H = M_2M_1 \times \frac{C_1B_1}{OC_1} = \alpha\left[\frac{(R\sin(225'))}{r}\right]. \tag{7.23}$$

Adding these differences to the Rsines etc. of $112\frac{1}{2}'$, we get the Rsines etc., of $1\frac{1}{2}$ arc-bits, that is $337\frac{1}{2}'$.

$$\begin{aligned} R\sin\left(337\tfrac{1}{2}'\right) = M_2P_2 &= M_1P_1 + M_2H \\ &= R\sin\left(112\tfrac{1}{2}'\right) + \alpha\left[\frac{(R\cos(225'))}{r}\right], \\ R\ \mathrm{vers}\left(337\tfrac{1}{2}'\right) = EP_2 &= EP_1 + M_1H \end{aligned} \tag{7.24}$$

$$\begin{aligned} &= R\,\mathrm{vers}\left(112\tfrac{1}{2}'\right) + \alpha\left[\frac{(R\sin(225'))}{r}\right], \quad (7.25)\\ R\cos\left(337\tfrac{1}{2}'\right) = OP_2 &= r - EP_2\\ &= R\cos\left(112\tfrac{1}{2}'\right) - \alpha\left[\frac{(R\sin(225'))}{r}\right]. \quad (7.26) \end{aligned}$$

The above values can now be used to calculate the second Rsines etc., by evaluating the Rsine-differences etc., between the arcs EC_2 and EC_1. For this purpose we note that the triangles OM_2P_2 and C_2C_1G are similar. Therefore,

$$C_2G = C_2C_1 \times \frac{OP_2}{OM_2} = \alpha\left[\frac{\left(R\cos\left(337\tfrac{1}{2}'\right)\right)}{r}\right], \quad (7.27)$$

$$C_1G = C_2C_1 \times \frac{M_2P_2}{OM_2} = \alpha\left[\frac{\left(R\sin\left(337\tfrac{1}{2}'\right)\right)}{r}\right]. \quad (7.28)$$

Adding these differences to the Rsines etc. of $225'$, we get the Rsines etc., of 2 arc-bits, that is $450'$.

$$\begin{aligned} R\sin(450') = C_2B_2 &= C_1B_1 + C_2G\\ &= R\sin\left(337\tfrac{1}{2}'\right) + \alpha\left[\frac{(R\cos(225'))}{r}\right], \quad (7.29)\\ R\,\mathrm{vers}(450') = EB_2 &= EB_1 + C_1G\\ &= R\,\mathrm{vers}\left(337\tfrac{1}{2}'\right) + \alpha\left[\frac{(R\sin(225'))}{r}\right], \quad (7.30)\\ R\cos(450') = OB_2 &= r - EB_2\\ &= R\cos\left(337\tfrac{1}{2}'\right) - \alpha\left[\frac{(R\cos(225'))}{r}\right]. \quad (7.31) \end{aligned}$$

Continuing this process we can calculate all the tabulated Rsine-differences, or equivalently, all the tabulated Rsines of multiples of the arc-bits $225'$. It is noted that in the process, we are also evaluating the Rsines of the mid-points of the arc-bits also, that is Rsines of odd multiples of $112\frac{1}{2}'$.[7]

[7]Note that the equations (7.18)–(7.31) are all exact relations. However, in order to evaluate the Rsines etc., we need to make a suitable approximation for the Rsine of the half-arc-bit $112\frac{1}{2}'$ and the full-chord α of the arc-bit $225'$, as given by equations (7.13) and (7.17).

7.4.3 Accurate Rsine and Rcosine at a desired point

When the desired arc is not a multiple of the arc-bit, then the Rsine etc. of the arc have to be obtained as corrections to the Rsines etc. of the nearest multiple of the arc-bit. The portion of the desired arc from its tip to the nearest tip of an arc-bit is called the *śiṣṭa-cāpa* or remainder arc. In Figure 7.6, let the tip of the i-th arc-bit C_i be closest to the tip of the desired arc EC. Then the *śiṣṭa-cāpa* is CC_i, which may be denoted by $\triangle$. The Rsine C_iB_i and Rcosine C_iK_i of the arc EC_i are tabulated Rsine and Rcosine values, which we assume are already calculated. To determine the Rsine, CB, and the Rcosine, CK, of the desired are EC, we need to calculate the corrections B_iB and K_iK. Let C_iK_i and CB meet at D.

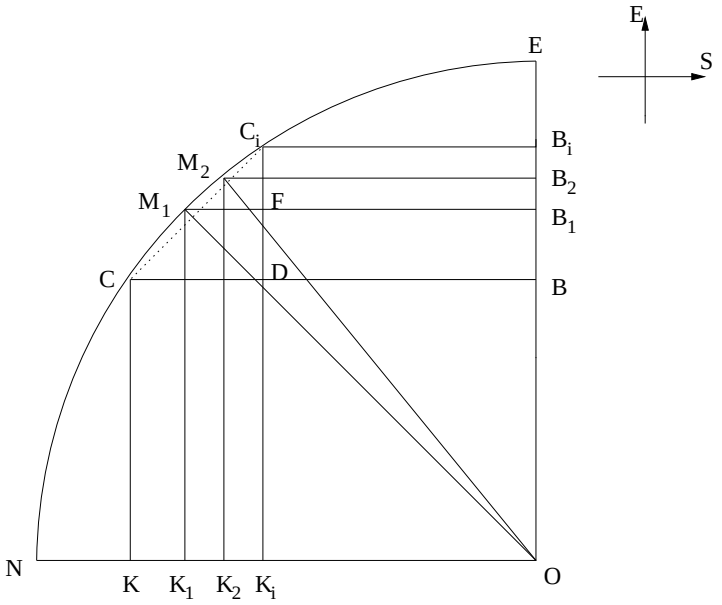


Figure 7.6: Rsine at a desired point.

If M_1 is the mid-point of the *śiṣṭa-cāpa* C_iC, and M_1B_1 and M_1K_1 are the Rsine and Rcosine of the arc EM_1, then we can follow the procedure outlined above, using the similarity of the triangles OM_1B_1 and CC_iD. We get

$$B_iB = C_iD = CC_i \times \frac{M_1B_1}{OM_1}\,, \tag{7.32}$$

$$K_iK = DC = CC_i \times \frac{OB_1}{OM_1}\,. \tag{7.33}$$

In the above, the chord CC_i of the *śiṣṭa-cāpa* can be approximated by the arc Δ itself.[8] But, we do not know the Rsine and Rcosine, M_1B_1 and OB_1, of the arc EM_1, which appear in (7.32), (7.33). For this purpose it is suggested that we may consider the mid-point M_2 of the arc C_iM_1, and the Rsine and Rcosine, M_2B_2 and M_2K_2, of the arc EM_2. If F is the point of intersection of C_iK_i and M_1B_1, then as before the triangles OM_2B_2 and M_1C_iF are similar and we get

$$B_iB_1 = C_iF = M_1C_i \times \frac{M_2B_2}{OM_2}\ , \tag{7.34}$$

$$K_iK_1 = FM_1 = M_1C_i \times \frac{OB_2}{OM_2}\ . \tag{7.35}$$

As before, the chord M_1C_i which appears in the above equations can be approximated by the corresponding arc $\frac{\Delta}{2}$. But again, we do not know the Rsine and Rcosine, M_2B_2 and OB_2, of the arc EM_2. At this stage, it is suggested that these may be approximated by the tabular Rsine and Rcosine, C_iB_i and OB_i, of the arc EC_i, because the portion of the arc left over, namely C_iM_2, may be considered negligible.[9]

From equations (7.32)–(7.35), we get

$$\begin{aligned} R\sin(EC) &= R\sin(EC_i + \Delta) = CB \\ &= C_iB_i + K_iK \\ &= R\sin EC_i + CC_i \times \frac{OB_1}{OM_1} \\ &= R\sin EC_i + CC_i \times \frac{(OB_i - B_iB_1)}{OM_1} \\ &= R\sin EC_i + \frac{CC_i}{r}R\cos EC_i - \frac{CC_i}{r}\left(\frac{M_1C_i}{r}\right)M_2B_2\ . \end{aligned} \tag{7.36}$$

[8]Note that the *śiṣṭa-cāpa* is always so chosen as to be less than or equal to half the arc-bit, or $112\frac{1}{2}'$, an arc which has already been assumed to be approximately equal to its Rsine, in (7.13), while calculating the tabulated Rsines.

[9]If, instead of going through all this, we had taken the arc C_iM_1 itself to be negligible and merely used equations (7.32) and (7.33) with the assumption that the Rsine and Rcosine, M_1B_i and OB_1, of the arc EM_1, may themselves be approximated by the tabulated Rsine and Rcosine, C_iB_i and OB_i of the arc EC_i, then we would obtain the first order approximations:

$$R\sin(EC_i + \Delta) \approx R\sin(EC_i) + \left(\frac{\Delta}{r}\right)R\cos(EC_i).$$

$$R\cos(EC_i + \Delta) \approx R\cos(EC_i) - \left(\frac{\Delta}{r}\right)R\sin(EC_i).$$

With the approximations that the chords CC_i and M_1C_i may be equated with Δ and $\frac{\Delta}{2}$, and the assumption that M_2B_2 may be approximated by the tabulated sine $R\sin(EC_i)$, we get

$$R\sin(EC_i+\Delta) \approx R\sin EC_i + \left(\frac{\Delta}{r}\right) R\cos EC_i - \left(\frac{1}{2}\right)\left(\frac{\Delta}{r}\right)^2 R\sin EC_i\,. \quad (7.37)$$

Similarly for the Rcosine of the arc EC we get,

$$R\cos(EC_i+\Delta) \approx R\cos EC_i - \left(\frac{\Delta}{r}\right) R\sin EC_i - \left(\frac{1}{2}\right)\left(\frac{\Delta}{r}\right)^2 R\cos EC_i\,. \quad (7.38)$$

These are the approximations to the Rsine and Rcosine of a desired arc as given in the rule *iṣṭadoḥkoṭidhanuṣoḥ*... (*Tantrasaṅgraha*, 2.10). It is also suggested that if this approximation is not good enough, then we may iterate the above process one more step by considering the midpoint of the arc C_iM_2 and so on.[10]

7.5 Computations of *Jyā* and *Śara* by *Saṅkalita*-s

7.5.1 First and second order differences of Rsines

So far we had considered the tabular Rsines (*paṭhita-jyā*) obtained by dividing the quadrant of a circle into 24 equal arc-bits. We shall now consider the

[10] If we do not approximate the Rsine and Rcosine, M_2B_2 and OB_2, of the arc EM_2 by the nearest tabulated Rsine and Rcosine, but derive them in terms of the Rsine and Rcosine at the midpoint M_2 of the arc C_iM_2 and then make a similar approximation, we will get the third order approximation,

$$R\sin(EC_i+\Delta) \approx R\sin EC_i + \left(\frac{\Delta}{r}\right) R\cos EC_i - \frac{1}{2}\left(\frac{\Delta}{r}\right)^2 R\sin EC_i + \frac{1}{2.4}\left(\frac{\Delta}{r}\right)^3 R\cos EC_i,$$

$$R\cos(EC_i+\Delta) \approx R\cos EC_i - \left(\frac{\Delta}{r}\right) R\sin EC_i - \frac{1}{2}\left(\frac{\Delta}{r}\right)^2 R\cos EC_i + \frac{1}{2.4}\left(\frac{\Delta}{r}\right)^3 R\sin EC_i.$$

In this context, it should be noted that while the first order approximation, given in the previous footnote, and the second order approximation, given in equations (7.37) and (7.38), appear like the first and second order terms in a Taylor's series expansion, there is no reason that the further terms should correspond to the terms in a Taylor's series expansion.

more general case of a given arc (*iṣṭa-cāpa*) of arc-length s, which is divided into n equal arc-bits. At this stage, the general results derived are equally applicable to the particular case of the quadrant divided into 24 equal parts. Later, we shall obtain approximate results for the Rsine and Rversine of an arbitrary arc, which become more and more accurate as n becomes larger and larger.

We first define the constituent Rsines (*piṇḍa-jyā*, or more accurately *piṇḍa-bhujā-jyā*) and the corresponding Rcosines and Rversines. If $s = r\theta$, then the j-th *piṇḍa-jyā*, B_j is given by

$$B_j = R\sin\left(\frac{js}{n}\right) = r\sin\left(\frac{j\theta}{n}\right) = r\sin\left(\frac{js}{rn}\right). \tag{7.39}$$

The corresponding *koṭi-jyā* K_j, and the *śara* S_j, are given by

$$K_j = R\cos\left(\frac{js}{n}\right) = r\cos\left(\frac{j\theta}{n}\right) = r\cos\left(\frac{js}{rn}\right), \tag{7.40}$$

$$S_j = R\,\mathrm{vers}\left(\frac{js}{n}\right) = r\left[1-\cos\left(\frac{j\theta}{n}\right)\right] = r\left[1-\cos\left(\frac{js}{rn}\right)\right]. \tag{7.41}$$

In Figure 7.7, EC is the desired arc of length s, and C_jC_{j+1} is the $(j+1)$-th arc bit. Then the arc $EC_j = \frac{js}{n}$, and its *piṇḍa-jyā* $B_j = C_jP_j$, and the corresponding *koṭi-jyā* and *śara* are $K_j = C_jT_j$, $S_j = EP_j$. Similarly we have $B_{j+1} = C_{j+1}P_{j+1}$, $K_{j+1} = C_{j+1}T_{j+1}$, and $S_{j+1} = EP_{j+1}$.

Let M_{j+1} be the mid-point of the arc-bit C_jC_{j+1} and similarly M_j the mid-point of the previous (j-th) arc-bit. We shall denote the *piṇḍa-jyā* of the arc EM_{j+1} as $B_{j+\frac{1}{2}}$ and clearly

$$B_{j+\frac{1}{2}} = M_{j+1}Q_{j+1}\,.$$

The corresponding *koṭi-jyā* and *śara* are

$$K_{j+\frac{1}{2}} = M_{j+1}U_{j+1} \quad \text{and} \quad S_{j+\frac{1}{2}} = EQ_{j+1}\,.$$

Similarly,

$$B_{j-\frac{1}{2}} = M_jQ_j,\ K_{j-\frac{1}{2}} = M_jU_j \ \text{ and } \ S_{j-\frac{1}{2}} = EQ_j\,.$$

The full-chord of the equal arc-bits $\frac{s}{n}$ may be denoted α as before. Thus,

$$C_jC_{j+1} = M_jM_{j+1} = \alpha.$$

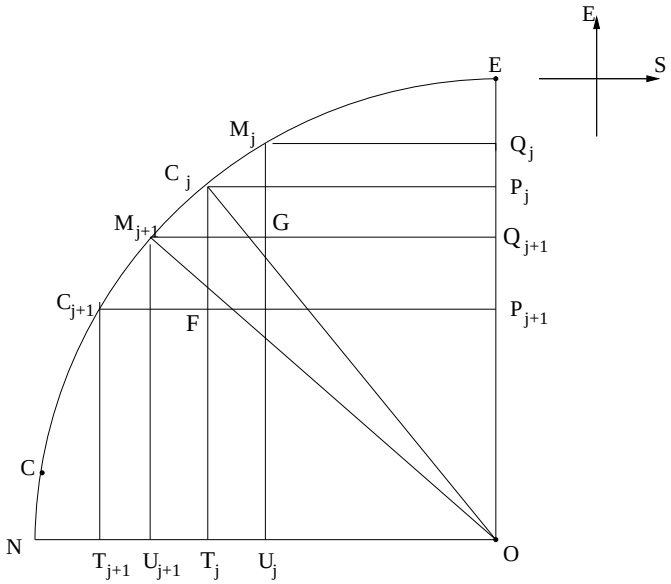


Figure 7.7: Computation of *Jyā* and *Śara* by *Saṅkalita*-s.

Let F be the intersection of C_jT_j and $C_{j+1}P_{j+1}$, and G of M_jU_j and $M_{j+1}Q_{j+1}$. As discussed before, the triangles $C_{j+1}FC_j$ and $OQ_{j+1}M_{j+1}$are similar, as their sides are mutually perpendicular. Thus we have

$$\frac{C_{j+1}C_j}{OM_{j+1}} = \frac{C_{j+1}F}{OQ_{j+1}} = \frac{FC_j}{Q_{j+1}M_{j+1}} \ . \tag{7.42}$$

Hence we obtain

$$B_{j+1} - B_j \ = \ \left(\frac{\alpha}{r}\right) K_{j+\frac{1}{2}}, \tag{7.43}$$

$$K_j - K_{j+1} \ = \ S_{j+1} - S_j = \left(\frac{\alpha}{r}\right) B_{j+\frac{1}{2}}. \tag{7.44}$$

Similarly, the triangles $M_{j+1}GM_j$ and OP_jC_j are similar and we get

$$\frac{M_{j+1}M_j}{OC_j} = \frac{M_{j+1}G}{OP_j} = \frac{GM_j}{P_jC_j}. \tag{7.45}$$

Thus we obtain

$$B_{j+\frac{1}{2}} \ - \ B_{j-\frac{1}{2}} = \left(\frac{\alpha}{r}\right) K_j, \tag{7.46}$$

$$K_{j-\frac{1}{2}} \ - \ K_{j+\frac{1}{2}} = S_{j+\frac{1}{2}} - S_{j-\frac{1}{2}} = \left(\frac{\alpha}{r}\right) B_j \ . \tag{7.47}$$

We define the Rsine-differences (*khaṇḍa-jyā*) Δ_j by

$$\Delta_j = B_j - B_{j-1} \,, \tag{7.48}$$

with the convention that $\Delta_1 = B_1$. From (7.43), we have

$$\Delta_j = \left(\frac{\alpha}{r}\right) K_{j-\frac{1}{2}} \,. \tag{7.49}$$

From (7.47) and (7.49), we also get the second order Rsine-difference (the differences of the Rsine-differences called *khaṇḍa-jyāntara*).[11]

$$\begin{aligned} \Delta_j - \Delta_{j+1} &= (B_j - B_{j-1}) - (B_{j+1} - B_j) \\ &= \left(\frac{\alpha}{r}\right)\left(K_{j-\frac{1}{2}} - K_{j+\frac{1}{2}}\right) \\ &= \left(\frac{\alpha}{r}\right)\left(S_{j+\frac{1}{2}} - S_{j-\frac{1}{2}}\right) \\ &= \left(\frac{\alpha}{r}\right)^2 B_j \,. \end{aligned} \tag{7.50}$$

Now, if the sum of the second-order Rsine-differences, are subtracted from the first Rsine-difference, then we get any desired Rsine-difference. That is

$$\Delta_1 - [(\Delta_1 - \Delta_2) + (\Delta_2 - \Delta_3) + \ldots + (\Delta_{j-1} - \Delta_j)] = \Delta_j \,. \tag{7.51}$$

From (7.50) and (7.51) we conclude that

$$\Delta_1 - \left(\frac{\alpha}{r}\right)^2 (B_1 + B_2 + \ldots + B_{j-1}) = \Delta_j \,. \tag{7.52}$$

7.5.2 Desired Rsines and Rversines from *Jyā-saṅkalita*

We can sum up the Rversine-differences (7.47), to obtain the *śara*, Rversine, at the midpoint of the last arc-bit as follows:

$$\begin{aligned} S_{n-\frac{1}{2}} - S_{\frac{1}{2}} &= \left(S_{n-\frac{1}{2}} - S_{n-\frac{3}{2}}\right) + \ldots\ldots \left(S_{\frac{3}{2}} - S_{\frac{1}{2}}\right) \\ &= \left(\frac{\alpha}{r}\right)(B_{n-1} + B_{n-2} + \ldots + B_1) \,. \end{aligned} \tag{7.53}$$

[11] This formula for the second order Rsine-difference, when applied to the tabulated Rsines obtained by dividing the quadrant of a circle into 24 parts, is nothing but the well-known relation given in the *Āryabhaṭīya* for the second order Rsine-differences. *Āryabhaṭīya* gives the approximate value $\left(\frac{1}{225'}\right)$ for the quantity $\left(\frac{\alpha}{r}\right)^2$ in (7.50); Nīlakaṇṭha (*Tantrasaṅgraha* 2.4) gives the more accurate value $233\frac{1}{2}'$ for the divisor, and Śaṅkara, in his commentary *Laghuvivṛtti*, refines it further to $233'32''$.

Using (7.50), the right hand side of (7.53) can also be expressed as a summation of the second order differences. From (7.52) and (7.53) it follows that the Rversine at the midpoint of the last arc-bit is also given by

$$\left(\frac{\alpha}{r}\right)\left(S_{n-\frac{1}{2}} - S_{\frac{1}{2}}\right) = (\Delta_1 - \Delta_n). \tag{7.54}$$

Now, since the first Rsine-difference $\Delta_1 = B_1$, any desired Rsine can be obtained by adding the Rsine-differences; these Rsine-differences have been obtained in (7.52). Before discussing the general result which is going to be obtained this way, a particular case, the eighth *piṇḍa-jyā*, is evaluated to elucidate the general result. From the definition (7.48) of the Rsine-differences, We have

$$B_8 = \Delta_8 + \Delta_7 + \ldots + \Delta_1 \,. \tag{7.55}$$

By using (7.52), equation (7.55) becomes

$$B_8 = 8\Delta_1 - \left(\frac{\alpha}{r}\right)^2 [(B_1 + B_2 + \ldots + B_7) + (B_1 + B_2 + \ldots + B_6) + \ldots + B_1]$$

$$= 8B_1 - \left(\frac{\alpha}{r}\right)^2 (B_7 + 2B_6 + \ldots + 7B_1). \tag{7.56}$$

In the same way, by making use of (7.52), the last *piṇḍa-jyā* can be expressed as follows:

$$B_n = \Delta_n + \Delta_{n-1} + \ldots + \Delta_1$$

$$= n\Delta_1 - \left(\frac{\alpha}{r}\right)^2 [(B_1 + B_2 \ldots + B_{n-1}) + (B_1 + B_2 \ldots + B_{n-2}) + \ldots + B_1]$$

$$= nB_1 - \left(\frac{\alpha}{r}\right)^2 [B_{n-1} + 2B_{n-2} + \ldots + (n-1)B_1] \,. \tag{7.57}$$

Equations (7.53) and (7.57) express the Rsines and Rversines around the tip of the desired arc in terms of the summations and repeated summations (*saṅkalita*) of the *piṇḍa-jyā*-s, Rsines at the tips of the arc-bits. By employing (7.53), we can also re-express the expression (7.57) for the last (*piṇḍa-jyā*) in the form

$$B_n = nB_1 - \left(\frac{\alpha}{r}\right)\left[S_{n-\frac{1}{2}} + S_{n-\frac{3}{2}} + \ldots + S_{\frac{1}{2}} - nS_{\frac{1}{2}}\right] \tag{7.58}$$

The results (7.46)–(7.58), obtained so far, involve no approximations. It is now shown how better and better approximations to the Rsine and Rversine

can be obtained by taking n to be very large or, equivalently, the arc-bit $\frac{s}{n}$ to be very small. Then, we can approximate the full-chord and the Rsine of the arc-bit by the length of the arc-bit $\frac{s}{n}$ itself. Also, as a first approximation, we can approximate the *piṇḍa-jyā*-s B_j in the equations (7.53), (7.57) or (7.58) by the corresponding arcs themselves. That is

$$B_j \approx \frac{js}{n} \,. \tag{7.59}$$

The result for the Rsine obtained this way is again used to obtain a better approximation for the *piṇḍa-jyā*-s B_j which is again substituted back into the equations (7.53) and (7.57) or (7.58) and thus by a process of iteration successive better approximations are obtained for the Rsine and Rversine. Now, once we take $B_j \approx \frac{js}{n}$, we will be led to estimate the sums and repeated sums of natural numbers (*ekādyekottara-saṅkalita*), when the number of terms is very large. This topic has been dealt with earlier in Chapter 6 and is briefly recalled here.

7.5.3 First, second, etc. repeated summations (*Ādya-dvitīyādi-saṅkalita*)

The first summation (*ādya-saṅkalita*) is that of the natural numbers

$$V_n^{(1)} = n + (n-1) + \ldots + 1. \tag{7.60}$$

This can be easily obtained from the series figure (*saṅkalita-kṣetra*, see Figure 7.8) to be

$$V_n^{(1)} = \frac{n(n+1)}{2} \,. \tag{7.61}$$

The second summation (*dvitīya-saṅkalita*) is given by

$$\begin{aligned} V_n^{(2)} &= V_n^{(1)} + V_{n-1}^{(1)} + \ldots + V_1^{(1)} \\ &= \frac{n(n+1)}{2} + \frac{(n-1)n}{2} + \ldots + \frac{(1.2)}{2} \,. \end{aligned} \tag{7.62}$$

This summation is stated to be

$$V_n^{(2)} = \frac{n(n+1)(n+2)}{1.2.3} \,. \tag{7.63}$$

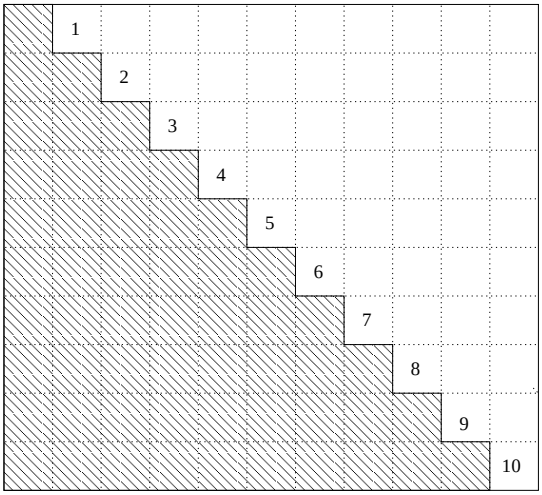


Figure 7.8: *Saṅkalita-kṣetra* for first summation.

Similarly the general k-th (repeated) summation is stated to be [12]

$$\begin{aligned} V_n^{(k)} &= V_n^{(k-1)} + V_{n-1}^{(k-1)} + \ldots + V_1^{(k-1)} \\ &= \frac{n(n+1)(n+2)\ldots(n+k)}{1.2.3.\ldots(k+1)} \,. \end{aligned} \tag{7.64}$$

What is needed for the evaluation of the Rsines and Rversines is the value of these summations for large n. They were derived earlier in Section 6.4.5 and are briefly recalled at this point. Clearly, for large n, the first summation is essentially half the square of the number terms n:

$$V_n^{(1)} \approx \frac{n^2}{1.2} \,. \tag{7.65}$$

Using the estimate (7.65) in (7.62) it follows that

$$V_n^{(2)} \approx \frac{n^2}{1.2} + \frac{(n-1)^2}{1.2} + \ldots \quad . \tag{7.66}$$

Thus, what we have is essentially half the summation of squares (*varga-saṅkalita*). When n is large, the sum of squares has been shown to be

[12] As we noted earlier, the result for the second (repeated) summation is given in *Āryabhaṭīya*, *Gaṇita* 21. The general result (7.64) for the k-th (repeated) summation is also given by Nārāyaṇa Paṇḍita (c.1350) in his *Gaṇitakaumudī*, 3.19.

one-third the cube of the number of terms and thus we get

$$V_n^{(2)} \approx \frac{n^3}{1.2.3} \,. \tag{7.67}$$

This way we can show that for large n, the k-th order (repeated) summation is given by

$$V_n^{(k)} \approx \frac{n^{k+1}}{1.2.3.\ldots(k+1)} \,. \tag{7.68}$$

7.5.4 Successive corrections to *Jyā* and *Śara*

We first recall the equations (7.53) and (7.57) which give the Rsine and Rversine near the tip of the desired arc in terms of *saṅkalita*-s of the *piṇḍa-jyā*-s:

$$S_{n-\frac{1}{2}} - S_{\frac{1}{2}} = \left(\frac{\alpha}{r}\right)(B_{n-1} + B_{n-2} + \ldots + B_1), \tag{7.69}$$

$$\begin{aligned} B_n &= nB_1 - \left(\frac{\alpha}{r}\right)^2 [(B_1 + B_2 + \ldots + B_{n-1}) \\ &\qquad + (B_1 + B_2 + \ldots + B_{n-2}) + \ldots + B_1]. \end{aligned} \tag{7.70}$$

As we noted earlier, these relations are exact. But now we shall show how better and better approximations to the Rsine and Rversine of any desired arc can be obtained by taking n to be very large or, equivalently, taking the arc-bit $\frac{s}{n}$ to be very small. Then both the full-chord α, and the first Rsine B_1 (the Rsine of the arc-bit), can be approximated by the arc-bit $\frac{s}{n}$ itself, and the Rversine $S_{n-\frac{1}{2}}$ can be taken as S_n and the Rversine $S_{\frac{1}{2}}$ may be treated as negligible. Thus the above relations become

$$S = S_n \approx \left(\frac{s}{nr}\right)(B_{n-1} + B_{n-2} + \ldots + B_1), \tag{7.71}$$

$$\begin{aligned} B = B_n &\approx s - \left(\frac{s}{nr}\right)^2 [(B_1 + B_2 + \ldots + B_{n-1}) \\ &\qquad + (B_1 + B_2 \ldots + B_{n-2}) + \ldots + B_1], \end{aligned} \tag{7.72}$$

where B and S are the Rsine and Rversine of the desired arc of length s and the results will be more accurate, larger the value of n.

It is possible to obtain another approximation for the Rsine by considering equation (7.58). When n is very large, it can be assumed that the Rversines at the mid-points are practically the same as the Rversines at the junctions of the arc-bits, and that the term involving $nS_{\frac{1}{2}}$ can be neglected.[13] We thus obtain

$$B = B_n \approx s - \left(\frac{s}{nr}\right)(S_n + S_{n-1} + \ldots + S_1). \tag{7.73}$$

Either of the equations (7.72) or (7.73) can be used in tandem with equation (7.71) to obtain successive corrections to Rsines and Rversines.

Now, as a first approximation, we take each *piṇḍa-jyā* B_j in (7.71) and (7.72) to be equal to the corresponding arc itself, that is

$$B_j \approx \frac{js}{n}\,. \tag{7.74}$$

Then we obtain for the Rversine

$$\begin{aligned} S &\approx \left(\frac{s}{nr}\right)\left[(n-1)\left(\frac{s}{n}\right) + (n-2)\left(\frac{s}{n}\right) + \ldots\right] \\ &= \left(\frac{1}{r}\right)\left(\frac{s}{n}\right)^2 [(n-1) + (n-2) + \ldots]. \end{aligned} \tag{7.75}$$

Now, from the estimate (7.65) for the *saṅkalita* on the right hand side of (7.75), we obtain

$$S \approx \left(\frac{1}{r}\right)\frac{s^2}{2}\,. \tag{7.76}$$

Equation (7.76) is the first *śara-saṃskāra*, correction to the Rversine. We now substitute our first approximation (7.74) to the *piṇḍa-jyā*-s B_j in (7.72), which gives the Rsine of the desired are as a second order repeated sum of the *piṇḍa-jyā*-s B_j. We then obtain

$$B \approx s - \left(\frac{1}{r}\right)^2\left(\frac{s}{n}\right)^3 [(1+2+\ldots+(n-1))+(1+2+\ldots(n-2))+\ldots]. \tag{7.77}$$

The second term in (7.77) is a *dvitīya-saṅkalita*, the second order repeated sum, and using the estimate (7.67), we obtain

$$B \approx s - \left(\frac{1}{r}\right)^2 \frac{s^3}{1.2.3}\,. \tag{7.78}$$

[13] Unlike the Rsines which are proportional to the arc-bit when the arc-bit is small, the Rversines are proportional to the square of the arc-bit and thus $nS_{\frac{1}{2}}$ can be neglected, unlike nB_1, when the arc-bit $\left(\frac{s}{n}\right)$ is very small.

Thus we see that the first correction obtained in (7.78) to the Rsine-arc-difference (*jyā-cāpāntara-saṃskāra*) is equal to the earlier correction to the Rversine (*śara-saṃskāra*) given in (7.76) multiplied by the arc and divided by the radius and 3.

Another way of deriving (7.78) is by employing the result (7.76) for the Rversines, for the *piṇḍa-śara-jyā*-s S_n, S_{n-1} etc. which appear in (7.73); then we obtain

$$B \approx s - \left(\frac{1}{r}\right)^2 \left(\frac{s}{n}\right)^3 \left(\frac{1}{2}\right) [n^2 + (n-1)^2 + \ldots]. \tag{7.79}$$

Now, the second term in (7.79) can either be estimated using (6.54) as a summation of squares (*varga-saṅkalita*), or by using (7.67) as a *dvitīya-saṅkalita*, second order (repeated) summation, because each individual term there has been obtained by doing a summation as in (7.75). The equation (7.79) then reduces to (7.78). This procedure can be adopted also for deriving the higher order corrections to the Rsine-arc-difference.

It is noted that the results (7.76) and (7.78) are only approximate (*prāyika*), since, instead of the *saṅkalita* of the *piṇḍa-jyā*-s in (7.71) and (7.72), we have only carried out *saṅkalita* of the arc-bits. Now that (7.78) gives a correction to the difference btween the Rsine and the arc (*jyā-cāpāntara-saṃskāra*), we can use that to correct the values of the *piṇḍa-jyā*-s and thus obtain the next corrections to the Rversine and Rsine.

Following (7.78), the *piṇḍa-jyā*-s may now be taken as

$$B_j \approx \frac{js}{n} - \left(\frac{1}{r}\right)^2 \left[\frac{\left(\frac{js}{n}\right)^3}{1.2.3}\right]. \tag{7.80}$$

If we introduce (7.80), in (7.71), we obtain

$$\begin{aligned} S &\approx \left(\frac{1}{r}\right)\left(\frac{s}{n}\right)^2 [(n-1) + (n-2) + \ldots] \\ &- \left(\frac{s}{nr}\right)\left(\frac{1}{r}\right)^2 \left(\frac{s}{n}\right)^3 \left(\frac{1}{1.2.3}\right) [(n-1)^3 + (n-2)^3 + \ldots]. \end{aligned} \tag{7.81}$$

The first term in (7.81) was already evaluated while deriving (7.76). The second term in (7.81) can either be estimated as a summation of cubes (*ghana-saṅkalita*), or as a *tritīya-saṅkalita*, third order (repeated) summation, because each individual term there has been obtained by doing a second-order

(repeated) summation. Hence, recollecting our earlier estimates for these *saṅkalita*-s, we get

$$S \approx \left(\frac{1}{r}\right) \frac{s^2}{1.2} - \left(\frac{1}{r}\right)^3 \frac{s^4}{1.2.3.4} \,. \tag{7.82}$$

Equation (7.82) gives a correction (*śara-saṃskāra*) to the earlier value (7.76) of the Rversine, which is nothing but the earlier correction to the Rsine-arc difference (*jyā-cāpāntara-saṃskāra*) given in (7.78) multiplied by the arc and divided by the radius and 4.

Again, if we use the corrected *piṇḍa-jyā*-s (7.80) in the expression (7.71) for the Rsine, we obtain

$$\begin{aligned} B &\approx s - \left(\frac{1}{r}\right)^2 \left(\frac{s}{n}\right)^3 [(1 + 2 + .. + (n-1)) + (1 + 2 + .. + (n-2)) + ..] \\ &\quad + \left(\frac{1}{r}\right)^4 \left(\frac{s}{n}\right)^5 \\ &\quad \times \left(\frac{1}{1.2.3}\right) \left[(1^3 + 2^3 + ... + (n-1)^3) + (1^3 + 2^3 + ... + (n-2)^3) + ..\right] \\ &\approx s - \left(\frac{1}{r}\right)^2 \frac{s^3}{1.2.3} + \left(\frac{1}{r}\right)^4 \frac{s^5}{1.2.3.4.5} \,, \end{aligned} \tag{7.83}$$

where, while evaluating the second order repeated summation in the third term, we have used the estimate (7.68) for the case of a fourth order repeated summation, as each individual term there has been obtained by a second order repeated summation as given in (7.77). Again, we may note that the second correction obtained in (7.83) to the Rsine-arc-difference (*jyā-cāpāntara-saṃskāra*) is equal to the earlier correction to the Rversine (*śara-saṃskāra*) given in (7.82) multitplied by the arc and divided by the radius and 5.

The above process can be repeated to obtain successive higher order corrections for the Rversine and Rsine: By first finding a correction (*jyā-cāpāntara-saṃskāra*) for the difference between the Rsine and the arc, using this correction to correct the *piṇḍa-jyā*-s B_j, and using them in equations (7.71) and (7.72) get the next correction (*śara-saṃskāra*) for the Rversines, and the next correction (*jyā-cāpāntara-saṃskāra*) for the Rsine-arc-difference itself, which is then employed to get further corrections iteratively.

7.5.5 Accurate computation of Rsines and Rversines, without using tables

The successive corrections to Rsine and Rversine, obtained as stated above, are going to be subtractive from the previous correction as they are based upon improving the estimate of the difference between the Rsine and the arc in our expression for the *piṇḍa-jyā*-s. We therefore obtain the following series of correction terms for the Rsine and Rversine:[14]

$$\begin{aligned} R\sin(s) &\approx s - \left(\frac{1}{r}\right)^2 \frac{s^3}{(1.2.3)} + \left(\frac{1}{r}\right)^4 \frac{s^5}{(1.2.3.4.5)} \\ &\qquad - \left(\frac{1}{r}\right)^6 \frac{s^7}{(1.2.3.4.5.7)} + \ldots, \qquad (7.84) \\ R\,\mathrm{vers}(s) &\approx \left(\frac{1}{r}\right)\frac{s^2}{2} - \left(\frac{1}{r}\right)^3 \frac{s^4}{(1.2.3.4)} + \left(\frac{1}{r}\right)^5 \frac{s^6}{(1.2.3.4.6)} - \ldots (7.85) \end{aligned}$$

These can also be re-expressed in the following form given by the rule *nihatya cāpavargeṇa*...(cited also in *Yuktidīpikā*, II. 440–443)

$$\begin{aligned} R\sin(s) &\approx s - s\,\frac{\left(\frac{s}{r}\right)^2}{(2^2+2)} + s\,\frac{\left(\frac{s}{r}\right)^4}{(2^2+2)(4^2+4)} - \ldots, \qquad (7.86) \\ R\,\mathrm{vers}(s) &\approx \frac{r\left(\frac{s}{r}\right)^2}{(1^2+1)} - \frac{r\left(\frac{s}{r}\right)^4}{(1^2+1)(3^2+3)} + \ldots \\ &\approx \frac{r\left(\frac{s}{r}\right)^2}{(2^2-2)} - \frac{r\left(\frac{s}{r}\right)^4}{(2^2-2)(4^2-4)} + \ldots\ . \qquad (7.87) \end{aligned}$$

The above expressions for *jyā* and *śara* can be employed to calculate them without using the tabular values, by using the sequence of numerical values

[14] Using the standard relations (7.4), (7.6) between *jyā* and sine and *śara* and versine, (7.84) and (7.85) reduce to the series

$$\begin{aligned} \sin\theta &= \theta - \frac{\theta^3}{(1.2.3)} + \frac{\theta^5}{(1.2.3.4.5)} - \frac{\theta^7}{(1.2.3.4.5.6.7)} + \cdots, \\ \mathrm{vers}\,\theta &= \frac{\theta^2}{(1.2)} - \frac{\theta^4}{(1.2.3.4)} + \frac{\theta^6}{(1.2..4.5.6)} - \cdots\ . \end{aligned}$$

given by the formulae, *vidvān* etc and *stena* etc.[15] This is indeed the method of deriving the required Rsine and Rversine without using the regular tabular values.

7.6 Accurate circumference from an approximate value

If d be the diameter of a circle and if we only know the approximate circumference c^*, a way to find a more accurate value of the circumference c, is the following. First evaluate the auxiliary Rsine value B^*, of the arc $\frac{c^*}{4}$ in a circle of radius d, as accurately as is desired. That is,

$$B^* = \frac{c^*}{4} - \left(\frac{1}{d}\right)^2 \frac{1}{1.2.3} \left(\frac{c^*}{4}\right)^3 + \left(\frac{1}{d}\right)^2 \frac{1}{1.2.3.4.5} \left(\frac{c^*}{4}\right)^5 \ldots \quad . \tag{7.88}$$

Let K^* be the corresponding Rcosine, namely

$$K^* = (d^2 - B^{*2})^{\frac{1}{2}}. \tag{7.89}$$

Let Δ be given by

$$\Delta = \left(\frac{B^{*2}}{2}\right)^{\frac{1}{2}} - \left(\frac{K^{*2}}{2}\right)^{\frac{1}{2}}. \tag{7.90}$$

Let δ be the arc corresponding to the Rsine value of Δ, which can be calculated by the relation

$$\Delta + \left(\frac{1}{d}\right)^2 \frac{1}{1.2.3} \Delta^3 \approx \delta. \tag{7.91}$$

[15]The formulae *vidvān* etc. and *stena* etc., are attributed to Mādhava by Nīlakaṇṭha in his *Āryabhaṭīya-bhāṣya, Gaṇitapāda*, 17 (see also, *Yuktidīpikā*, II. 437–438). If the quadrant of a circle is assigned the measure $q = 5400'$, then for a given arc s, the corresponding Rsine given by (7.86) can be expressed in the form:

$$R\sin(s) \approx s - \left(\frac{s}{q}\right)^3 \left(u_3 - \left(\frac{s}{q}\right)^2 \left(u_5 - \left(\frac{s}{q}\right)^2 \left(u_7 - \left(\frac{s}{q}\right)^2 \left(u_9 - \left(\frac{s}{q}\right)^2 u_{11}\right)\right)\right)\right),$$

where, the values of u_{11}, u_9, u_7, u_5, and u_3 are given by the formulae *vidvān* etc., as follows: $u_{11} = 44'''$, $u_9 = 33''06'''$, $u_7 = 16'05''41'''$, $u_5 = 273'57''47'''$ and $u_3 = 2220'39''40'''$. The above expression gives the values of Rsine of any arc accurately up to the thirds. Similarly, the formulae *stena* etc., are used to calculate the Rversine of any arc accurately up to the thirds. Mādhava has also given the tabulated sine values (for arcs in multiples of $225'$) accurately to the thirds in the rule *śreṣṭhaṃ nāma variṣṭhānāṃ...*(cited by Nīlakaṇṭha in his *Āryabhaṭīyabhāṣya*, *Gaṇitapāda*, 12).

Then, an accurate circumference is given by

$$c = 4\left[\left(\frac{c^*}{4}\right) + \delta\right]. \tag{7.92}$$

The rationale for this procedure is also explained. If c^* is the accurate circumference itself, then B^{*2} and K^{*2} will be the squares of the Rsine and Rcosine of one-eighth the circumference of a circle with radius (and not diameter) d, and hence will be equal to $\frac{d^2}{2}$. Hence (by the *jīve-paraspara-nyāya* to be stated shortly), Δ is nothing but the Rsine of the difference of one-eighth the true circumference and rough circumference of that circle. Therefore four times the corresponding arc will give the correction to the approximate value of the circumference c^*.

7.7 Square of Rsine

The square of the Rsine of an arc is given by the rule *nihatya cāpavargeṇa ...* (*Yuktidīpikā* II. 455–456)

$$R\sin^2(s) = s^2 - s^2\frac{\left(\frac{s}{r}\right)^2}{\left(2^2 - \frac{2}{2}\right)} + s^2\frac{\left(\frac{s}{r}\right)^4}{\left(2^2 - \frac{2}{2}\right)\left(3^2 - \frac{3}{2}\right)} - \ldots \ . \tag{7.93}$$

The values of this can also be found by using the formulae *śaurirjayati...*(cited also in *Yuktidīpikā* II. 457–458)

7.8 Derivation of Rsines from *Jīve-paraspara-nyāya*

7.8.1 *Jīve-paraspara-nyāya*

The *jīve-paraspara-nyāya* is the famous rule of Mādhava (cited for instance, in *Tantrasaṅgraha* 2.12), which corresponds to the following result. If s_1 and s_2 are two arcs then

$$R\sin(s_1 \pm s_2) = \left(\frac{1}{r}\right)[R\sin(s_1)\ R\cos(s_2) \pm R\cos(s_1)\ R\sin(s_2)]. \tag{7.94}$$

The proof of this is illustrated for the case when s_1 is the arc composed of the first and the second arc-bits and s_2 is the third arc-bit, when a quadrant is divided as usual into 24 equal arc-bits. In the figure 7.9 below (which is not to scale), s_1 is the arc EA_2 and s_2 is the arc A_2A_3. Join the radius OA_2 and let it meet the full-chord of the arc A_1A_3 at its midpoint L. Now, $R\sin(s_1) = A_2B_2$, $R\cos(s_1) = A_2S_2$, and since A_3L is the half-chord of the arc A_1A_3 which is double s_2, we have $R\sin(s_2) = A_3L$ and $R\cos(s_2) = OL$. From L draw LC parallel to the north-south line LD parallel to the east-west line. Let the third Rsine A_3B_3 meet LD at P and let the first Rcosine A_1S_1 meet LC at Q.

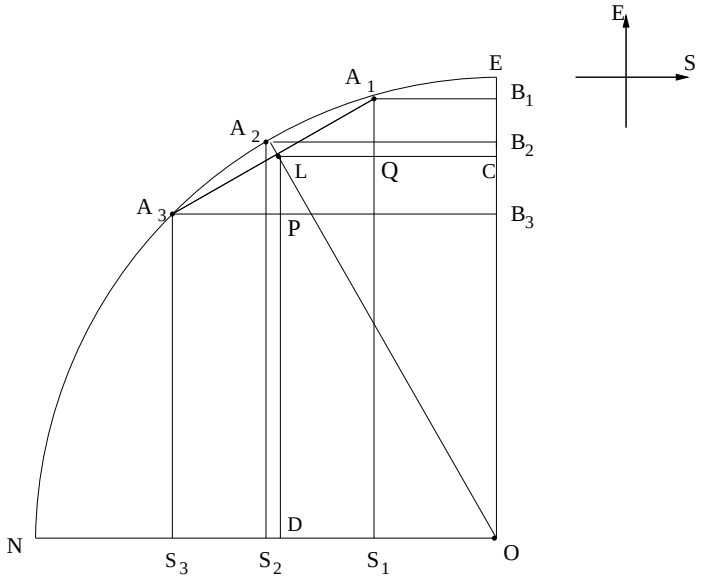


Figure 7.9: *Jīve-paraspara-nyāya.*

Now, the triangles A_3PL and A_2S_2O are similar as the sides are mutually perpendicular. Hence,

$$\frac{A_3P}{A_2S_2} = \frac{A_3L}{A_2O} = \frac{PL}{S_2O}\,. \tag{7.95}$$

Hence, we get

$$\left(\frac{1}{r}\right) R\sin(s_1)R\sin(s_2) \;=\; PL, \tag{7.96}$$

$$\left(\frac{1}{r}\right) R\cos(s_1)R\sin(s_2) \;=\; PA_3. \tag{7.97}$$

Then, triangles OLC and OA_2B_2 are similar. Hence,

$$\frac{LC}{A_2B_2} = \frac{OL}{OA_2}\,. \tag{7.98}$$

Therefore,

$$\left(\frac{1}{r}\right) R\sin(s_1)R\cos(s_2) = LC. \tag{7.99}$$

Then again, triangles OLD and OA_2S_2 are similar. Hence,

$$\frac{LD}{A_2S_2} = \frac{OL}{OA_2}\,. \tag{7.100}$$

Hence,

$$\left(\frac{1}{r}\right) R\cos(s_2)R\cos(s_1) = LD. \tag{7.101}$$

Now we have the relations

$$\begin{aligned} LC &+ A_3P = A_3B_3, \\ LC &- A_3P = A_1B_1. \end{aligned}$$

These correspond to the relations

$$R\sin(s_1 \pm s_2) = \left(\frac{1}{r}\right)[R\sin(s_1)\ R\cos(s_2) \pm R\cos(s_1)R\sin(s_2)]. \tag{7.102}$$

Similarly, we have

$$\begin{aligned} LD &- PL = A_3S_3, \\ LD &+ PL = A_1S_1. \end{aligned}$$

These correspond to the relations

$$R\cos(s_1 \pm s_2) = \left(\frac{1}{r}\right)[R\cos(s_1)\ R\cos(s_2) - (\pm)R\sin(s_1)R\sin(s_2)]. \tag{7.103}$$

These relations (7.102) and (7.103) can be used to calculate various tabulated Rsines, starting from the first Rsine. They can also be used to find the Rsine and Rcosine of a desired arc from the nearest tabular values and the Rsine and Rcosine of the remaining arc (*śiṣṭa-cāpa*).

7.8.2 *Jīve-paraspara-nyāya*: An alternative proof

The second proof is also for the same case and we can refer to the same figure 7.9. We start with the evaluation of A_3P and PL which has already been done in equations (7.96) and (7.97). Now consider the triangle A_3LB_3. Here, it has already been noted that A_3L is $R\sin(s_2)$. It is noted that LB_3 is $R\sin(s_1)$.[16] The perpendicular LP divides the base A_3B_3 into the two intercepts (*ābādhā*-s), A_3P and B_3P. It is well known that these base-intercepts are equal to the difference of the squares of the corresponding side and the perpendicular. We have already evaluated one intercept A_3P and the perpendicular LP in equations (7.96) and (7.97). The other intercept

$$\begin{aligned} B_3P^2 &= B_3L^2 - LP^2 \\ &= R\sin^2(s_1) - \left(\frac{1}{r}\right)^2 R\sin^2(s_1)R\sin^2(s_2). \end{aligned} \tag{7.104}$$

Therefore,

$$B_3P = \left(\frac{1}{r}\right) R\sin(s_1)R\cos(s_2). \tag{7.105}$$

The expression (7.97) for A_3P can also be derived in this way. Now using (7.97) and (7.105), we get

$$\begin{aligned} R\sin(s_1+s_2) &= A_3B_3 = B_3P + A_3P \\ &= \left(\frac{1}{r}\right)[R\sin(s_1)R\cos(s_2) + R\cos(s_1)R\sin(s_2)]. \end{aligned} \tag{7.106}$$

7.9 Principle of the area of a triangle

With a view to arrive at a method for deriving the Rsines without using the radius, the principles of the area of a triangle and a cyclic quandrilateral are dealt with. First the area of a triangle is discussed:

Consider a general scalene triangle (*viṣama-tryaśra*) ABC, with vertex A and base BC as in Figure 7.10. Draw the perpendicular AD from the vertex

[16] This can be shown by noting that LB_3 is half the full-chord of an arc which is double of the arc A_1A_3 which is equal to s_1. For this, produce A_3B_3 to meet the circle at D_1 (not shown in the figure). Then, in the triangle $A_3A_1D_1$, L is the midpoint of A_3A_1 and B_3 the midpoint of A_3D_1. Hence LB_3 is half the chord A_1D_1, which is the full chord of an arc equal to four arc-bits, or an arc which is double the arc A_1A_3.

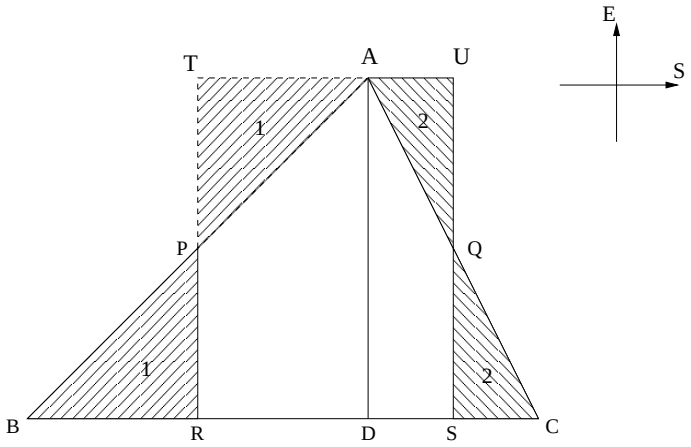


Figure 7.10: Area of a triangle.

to the base. Then the two base-intercepts BD, DC and the attitude AD can be evaluated in terms of the sides as follows.

$$BD^2 = AB^2 - AD^2, \tag{7.107}$$

$$CD^2 = AC^2 - AD^2. \tag{7.108}$$

Hence

$$BD^2 - CD^2 = AB^2 - AC^2. \tag{7.109}$$

Since $BC = BD + CD$, we get (assuming $AB > AC$, which implies $BD > CD$)

$$BD = \frac{1}{2}\left[BC + \frac{AB^2 - AC^2}{BC}\right], \tag{7.110}$$

$$CD = \frac{1}{2}\left[BC - \frac{AB^2 - AC^2}{BC}\right], \tag{7.111}$$

and

$$AD = (AB^2 - BD^2)^{\frac{1}{2}}. \tag{7.112}$$

The area of the triangle is half the base BC multiplied by the altitude AD. This can be seen by cutting the triangle along the lines joining the midpoints of the sides with the midpoints of the base-segments and reconstituting them into a rectangle as shown in Figure 7.10.

7.10 Diagonals of a cyclic quadrilateral

Before proceeding to evaluate the area of a cyclic quadrilateral, we first obtain expression for its diagonals in terms of its sides. In Figure 7.11 we consider a cyclic quadrilateral $ABCD$, such that its base AB on the western side is the largest side and the face CD on the east in the smallest. The side BC to the south is smaller than AB, but larger than the other side DA to the north. The sides are full-chords of the associated arcs, and the diagonals AC and BD are full-chords of arcs which are sums of the arcs associated with adjacent sides. First we shall obtain a result about such chords.

7.10.1 Product of two full-chords is equal to the difference in the squares of the full-chords associated with half the sum and difference of the arcs

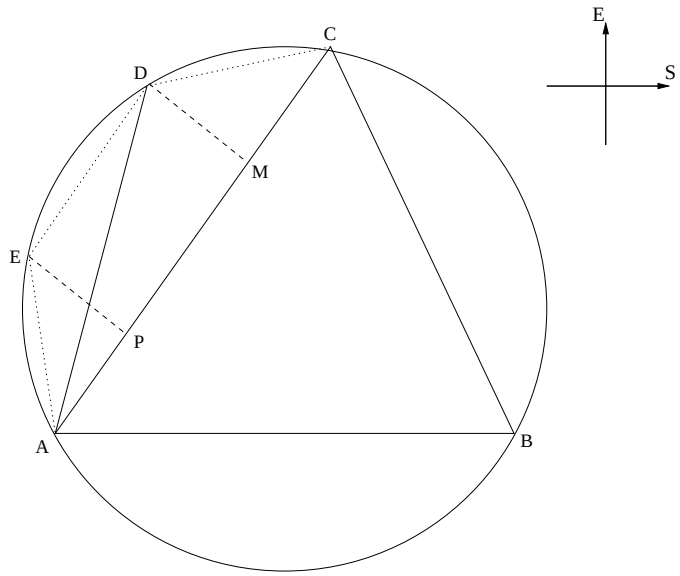


Figure 7.11: Cyclic quadrilateral.

In Figure 7.11, if DM is the perpendicular from the vertex D of the cyclic quadrilateral $ABCD$ onto the diagonal AC, then in the triangle ADC, the difference in the squares of the sides $AD,\ DC$ is equal to the difference in

the square of the base segments ($\bar{a}b\bar{a}dh\bar{a}$)

$$AD^2 - DC^2 = AM^2 - MC^2. \tag{7.113}$$

Hence,

$$\begin{aligned} AD^2 - DC^2 &= (AM + MC)(AM - MC) \\ &= AC(AM - MC). \end{aligned} \tag{7.114}$$

Now, $AM + MC = AC$ is the full-chord associated with sum of the arcs associated with the chords AD and DC. Similarly, $AM - MC$ is the chord associated with the difference of the arcs associated with AD and DC. To see this, mark off E on the circle, such that arc AE is equal to the arc CD. Then, chord AE is equal to the chord CD and hence, DE is parallel to AC and $ACDE$ is a trapezium and $DE = PM = AM - MC$.

Hence $AM - MC$ is the chord associated with the arc DE which is the difference between the arc AD and arc CD. If we denote these arcs by s_1, s_2, then we have shown[17] (with our convention that $s_1 > s_2$)

$$\text{Chord}^2(s_1) - \text{Chord}^2(s_2) = \text{Chord}(s_1 + s_2)\text{Chord}(s_1 - s_2). \tag{7.115}$$

Equation (7.115) is also equivalent to the following:

$$\text{Chord}(s_1)\text{Chord}(s_2) = \text{Chord}^2\left[\frac{(s_1 + s_2)}{2}\right] - \text{Chord}^2\left[\frac{(s_1 - s_2)}{2}\right]. \tag{7.116}$$

7.10.2 Sum of the products of the two pairs of sides associated with a diagonal is equal to the product of the diagonal with the third diagonal

In Figure 7.12 we consider the cyclic quadrilateral $ABCD$ with the specifications mentioned above, namely that, the base AB on the western side is the largest side and the face CD on the east is the smallest side and the side BC to the south is smaller than AB, but larger than the other side DA to the north.

[17] Relations (7.115) and (7.116) are equivalent to the trigonometric relations:

$$\begin{aligned} \sin^2(\theta_1) - \sin^2(\theta_2) &= \sin(\theta_1 + \theta_2)\sin(\theta_1 - \theta_2), \\ \sin(\theta_1)sin(\theta_2) &= \sin^2\left[\frac{(\theta_1 + \theta_2)}{2}\right] - \sin^2\left[\frac{(\theta_1 - \theta_2)}{2}\right]. \end{aligned}$$

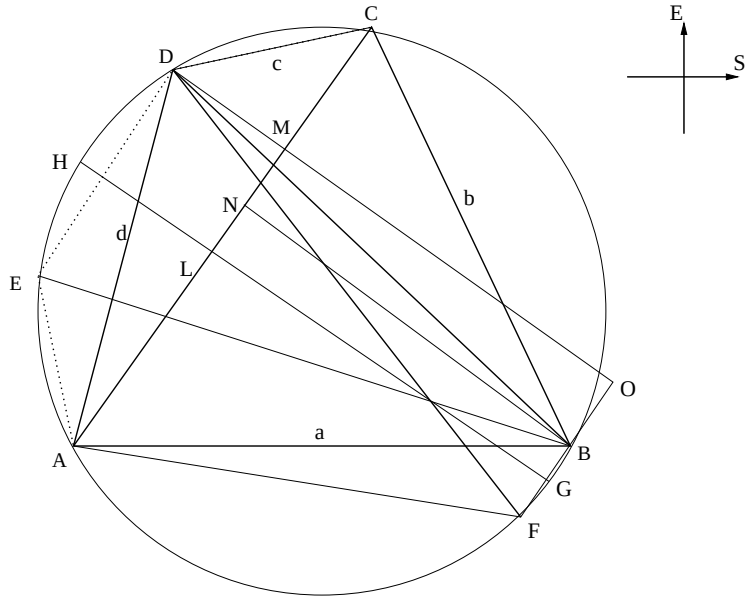


Figure 7.12: Diagonals of a cyclic quadrilateral.

In the arc AC, mark the point E such that arc AE is equal to the arc of the face DC . Then the chord ED is parallel to AC and

$$\text{Arc}(ED) = \text{Arc}(AD) - \text{Arc}(DC). \tag{7.117}$$

In the same way mark the point F in the arc AB such that arc AF is equal to the arc BC. Then the chord FB is parallel to AC and

$$\text{Arc}(FB) = \text{Arc}(AB) - \text{Arc}(BC). \tag{7.118}$$

Let H be the midpoint of arc ED and HG be the diameter of the circle. Then, we have

$$\begin{aligned} \text{Arc}(AH) &= \text{Arc}(AE) + \text{Arc}(EH) \\ &= \text{Arc}(CD) + \text{Arc}(DH) \\ &= \text{Arc}(HC). \end{aligned} \tag{7.119}$$

Also, since HG is the diameter,

$$\text{Arc}(HA) + \text{Arc}(AG) = \text{Arc}(HC) + \text{Arc}(CG). \tag{7.120}$$

From (7.119) and (7.120) it follows that.

$$\text{Arc}(AG) = \text{Arc}(GC). \tag{7.121}$$

Since, arcs AF and BC are equal, it follows that G is the midpoint of FB, and

$$\text{Arc}(FG) = \text{Arc}(GB) = \frac{1}{2}[\text{Arc}(AB) - \text{Arc}(BC)]. \tag{7.122}$$

Now BD may be termed the first diagonal of the quadrilateral and AC the second diagonal. As mentioned earlier, these are full-chords associated with sums of the arcs associated with different pairs of adjacent sides. There is one more "diagonal" which is the full-chord associated with the sum of the arcs associated with the opposite sides. This can also be obtained, say by interchanging the sides AB and BC in the quadrilateral, which leads to the quadrilateral $AFCD$, which has the diagonal DF. This is the "third diagonal" of the original quadrilateral $ABCD$, and

$$\text{Arc}(DF) = \text{Arc}(DC) + \text{Arc}(CF) = \text{Arc}(DC) + \text{Arc}(AB). \tag{7.123}$$

Now, if we instead interchange the sides AD and DC, then we get the quadrilateral $ABCE$, which has the diagonal EB. But

$$\begin{aligned} \text{Arc}(EB) &= \text{Arc}(EA) + \text{Arc}(AB) \\ &= \text{Arc}(DC) + \text{Arc}(AB) \\ &= \text{Arc}(DF). \end{aligned} \tag{7.124}$$

Hence, we are led to the same third diagonal $EB = DF$. It is also noted that since, we have exhausted all the possibilities of choosing pairs from a given set of four arcs, there are no more diagonals possible.

Now we consider the sum $AD.DC + AB.BC$. This is said to be the product of the sides associated with the first diagonal DB (*ādyakarṇāśrita-bhujāghātaikya*). From the earlier result (7.116) on the product of two full-chords being equal to the difference in the squares of the full-chords associated with half their sum and half their difference, we get

$$\begin{aligned} AD \times DC &= \text{Chord}^2\left[\frac{1}{2}(\text{Arc}(AD) + \text{Arc}(DC))\right] \\ &\quad - \text{Chord}^2\left[\frac{1}{2}(\text{Arc}(AD) - \text{Arc}(DC))\right], \end{aligned} \tag{7.125}$$

$$\begin{aligned} AB \times BC &= \text{Chord}^2\left[\frac{1}{2}(\text{Arc}(AB) + \text{Arc}(BC))\right] \\ &\quad - \text{Chord}^2\left[\frac{1}{2}(\text{Arc}(AB) - \text{Arc}(BC))\right]. \end{aligned} \tag{7.126}$$

Now, since the sum of the arcs AB, BC, CD and DA is the circumference, half the sum of the arcs AB and BC will be the complement of half the sum of the arcs CD and DA and hence the first terms in equations (7.125), (7.126) are the *bhujā* and *koṭi* of each other in a semicircle. Thus, if d is the diameter of the circle

$$\begin{aligned}\text{Chord}^2\left[\frac{1}{2}(\text{Arc}(AD)+\text{Arc}(DC))\right]\\ +\text{Chord}^2\left[\frac{1}{2}(\text{Arc}(AB)+\text{Arc}(BC))\right]=d^2.\end{aligned}\tag{7.127}$$

Now, the diameter is the full-chord associated with arc GH, and half the difference of the arcs AD and DC is given by the arcs HE and HD. Hence, applying (7.115), we get

$$d^2-\text{Chord}^2\left[\frac{1}{2}[\text{Arc}(AD)-\text{Arc}(DC)]\right]=\text{Chord}(GE).\text{Chord}(GD).\tag{7.128}$$

Now, since the arcs GE and GD are obtained by subtracting equal amounts from the diameter, the corresponding chords are equal and hence,

$$d^2-\text{Chord}^2\left[\frac{1}{2}[\text{Arc}(AD)-\text{Arc}(DC)]\right]=GD^2.\tag{7.129}$$

Finally, if we note that half the difference of the arcs AB and BC is given by the arc GB, we get from (7.125)–(7.129) that

$$AD\times DC+AB\times BC=GD^2-GB^2.\tag{7.130}$$

On using the relation (7.115), equation (7.130) becomes

$$\begin{aligned}AD\times DC+AB\times BC &= \text{Chord}[\text{Arc}(GD)+\text{Arc}(GB)]\times\\ &\quad\ \text{Chord}[\text{Arc}(GD)-\text{Arc}(GB)]\\ &= \text{Chord}[\text{Arc}(GD)+\text{Arc}(GF)]\times\\ &\quad\ \text{Chord}[\text{Arc}(GD)-\text{Arc}(GB)]\\ &= DF\times DB.\end{aligned}\tag{7.131}$$

Thus we have shown that the sum of the products of the two pairs of sides associated with the first diagonal is equal to the product of the first diagonal with the third diagonal.

7.10.3 The area of a cyclic quadrilateral is equal to the product of the three diagonals divided by twice the circum-diameter

In Figure 7.12, consider the second diagonal AC as the base of the two triangles ABC and ADC. Now from the vertex D draw the perpendicular DM to AC and produce it to meet FB at O, and from the vertex B draw the perpendicular BN to AC. Since we have already noted that the chords DE and FB are both parallel to AC, it follows that BN and DM are also parallel and the sum of the perpendiculars is given by

$$DM + BN = DM + MO = DO. \tag{7.132}$$

The area of the cyclic quadrilateral $ABCD$ is the sum of the areas of the two triangles ABC and ACD, and is hence given by

$$A = \frac{1}{2}AC(DM + BN) = \frac{1}{2}AC \times DO. \tag{7.133}$$

We now employ the result (which shall be proved later in Section 7.13) that the product of two sides of triangle divided by the diameter is equal to the altitude drawn to the third side. Using this in the triangle DBF, we get the altitude DO to be

$$DO = \frac{DB \times DF}{d}\ . \tag{7.134}$$

From (7.133) and (7.134) we obtain

$$A = \frac{AC \times DB \times DF}{2d}\ . \tag{7.135}$$

Thus, the area of a cyclic quadrilateral is given by the product of the three diagonals divided by twice the circum-diameter.

7.10.4 Derivation of the *Karṇas* (diagonals)

We have already shown in (7.131) that the sum of the products of the two pairs of sides related to the first diagonal is equal to the product of the first and third diagonals. Similarly, we can show that the sum of the products of the two pairs of sides associated with the second diagonal is the product of the second and the third diagonals. That is

$$AB \times AD + DC \times BC = DF \times AC. \tag{7.136}$$

Finally, it can be shown in the same way that the sum of the products of the opposite sides is equal to the product of the first and second diagonals. That is

$$AB \times CD + AD \times BC = BD \times AC. \tag{7.137}$$

From the equations (7.131), (7.136) and (7.137), we can obtain the three diagonals in terms of the sides of the cyclic quadrilaterals. We shall adopt the notation: $AB = a,\ BC = b,\ CD = c$, and $DA = d$ (not to be confused with the diameter) for the sides and $DB = x,\ AC = y$ and $DF = z$ for the diagonals. Then we obtain

$$x = \left[(ab + cd)\frac{(ac + bd)}{(ad + bc)}\right]^{\frac{1}{2}}, \tag{7.138}$$

$$y = \left[(ac + bd)\frac{(ad + bc)}{(ab + cd)}\right]^{\frac{1}{2}}, \tag{7.139}$$

$$z = \left[(ab + cd)\frac{(ad + bc)}{(ac + bd)}\right]^{\frac{1}{2}}. \tag{7.140}$$

The expression (7.135) for the area of the cyclic quadrilateral becomes

$$A = \frac{x.y.z}{2d}\ . \tag{7.141}$$

7.11 Cyclic quadrilateral and *Jīve-paraspara-nyāya*

It is now shown that the result (7.137) that the sum of the product of opposite sides is equal to the product of the diagonals can be viewed as a result for Rsines, and then it will be the same as the *jīve-paraspara-nyāya* discussed earlier. First the proof of the *jīve-paraspara-nyāya* given in Section 7.8 is recalled.

Now, to understand that the *jīve-paraspara-nyāya* is equivalent to the result (7.137), we shall consider in Figure 7.13, the quadrant of a circle EA divided into a number of equal arc-bits as before. A_2A_4 is the full-chord of the third and fourth arc-bits together and the radius OA_3 bisects it perpendicularly at L. Let A_2M_2 be the second sine. Since both L and M_2 lie on semi-circles with OA_2 as the diameter, OLA_2M_2 is a cyclic quadrilateral with the radius OA_2 as a diagonal. The side A_2L, being a half-chord of two arc-bits, is nothing but the Rsine of one arc-bit. Then OL will be the corresponding

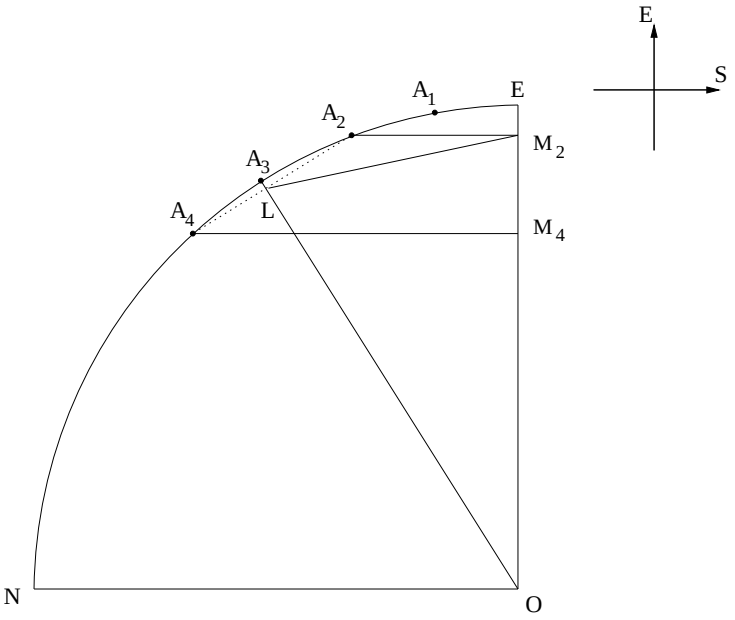


Figure 7.13: Cyclic quadrilateral and *Jīve-paraspara-nyāya*.

Rcosine. As already noted A_2M_2 is the Rsine of two arc-bits, and then OM_2 will be the corresponding Rcosine. It is said that the other diagonal LM_2 will be the Rsine of three arc-bits.[18]

If we now denote one arc-bit by s_1 and two arc-bits by s_2, then the *jīve-paraspara-nyāya*, (7.102),

$$R\sin(s_1)R\cos(s_2) + R\cos(s_1)R\sin(s_2) = r\ R\sin(s_1 + s_2), \qquad (7.142)$$

becomes

$$LA_2 \times OM_2 + OL \times A_2M_2 = OA_2 \times LM_2\ . \qquad (7.143)$$

This is the same as the relation (7.137) for the sum of the product of opposite sides of the cyclic quadrilateral OLA_2M_2.

[18]This can be proved by showing that LM_2 is equal to the half-chord of an arc which is double the arc EA_3. The procedure is similar to the one outlined in fn.16.

7.12 Derivation of tabular Rsines without using the radius

If $B_1, B_2 \ldots$, are the various tabulated Rsines when the quadrant is divided into a number of equal arc-bits, then we can use the equation (7.115) for the difference in the squares of chords of two arcs to obtain

$$B_2^2 - B_1^2 = B_3 B_1, \tag{7.144}$$

$$B_3^2 - B_1^2 = B_4 B_2 \, . \tag{7.145}$$

In general, we have

$$B_{j+1} = \frac{(B_j^2 - B_1^2)}{B_{j-1}} \, . \tag{7.146}$$

This can be used to calculate the tabulated Rsines without using the radius.

7.13 Altitude and circum-diameter of a triangle

Now is proved the result that the product of two sides of a triangle divided by its circum-diameter is equal to the altitude on the third side, which was used to obtain the expression (7.135) for the area of the cyclic quadrilateral in terms of its diagonals. In Figure 7.14, A_4B_4, A_6B_6 and $A_{10}B_{10}$ are the full-chords associated with the fourth, sixth and tenth tabulated Rsines; A_2 is the tip of the second arc-bit. Then A_2B_{10} is the full-chord associated with 12 arc-bits and A_2A_{10} is the full chord associated with 8 arc-bits. In the triangle, $A_2A_{10}B_{10}$, draw the altitude A_2P to the base $A_{10}B_{10}$.

Draw the chords A_6C_6 and B_4D_4 parallel to the east-west line and hence perpendicular to the full-chord $A_{10}B_{10}$ associated with the tenth Rsine. Draw also the diameters A_6OD_6 and B_4OC_4. These diameters intersect the full-chords A_2A_{10} and A_2B_{10} perpendicularly. Now the triangles $A_2A_{10}P$ and $D_6A_6C_6$ are similar as their sides are mutually perpendicular. Hence

$$\frac{A_2P}{D_6C_6} = \frac{A_2A_{10}}{D_6A_6} = \frac{A_{10}P}{A_6C_6} \, . \tag{7.147}$$

Hence

$$A_2P = A_2A_{10} \times \frac{D_6C_6}{D_6A_6} \, . \tag{7.148}$$

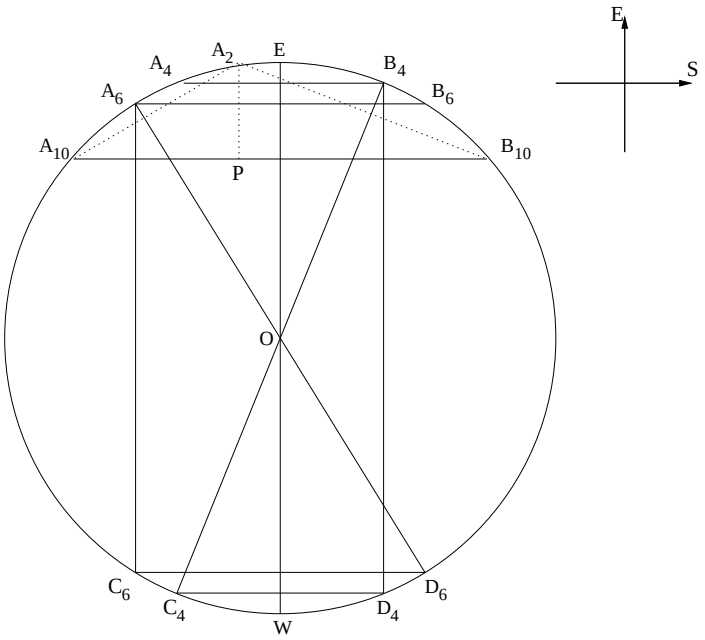


Figure 7.14: Altitude and circum-diameter of a triangle.

If we note that D_6A_6 is the diameter and D_6C_6, the full-chord associated with 12 arc-bits, is equal to A_2B_{10}, then we obtain

$$A_2P = A_2A_{10} \times \frac{A_2B_{10}}{d} \,. \tag{7.149}$$

Thus we have proved that in the triangle $A_2A_{10}B_{10}$, the product of the sides A_2A_{10} and A_2B_{10} divided by the diameter is equal to the altitude A_2P to the third side $A_{10}B_{10}$.

Now, we may also note that the triangle $A_2B_{10}P$ and $C_4B_4D_4$ are similar as their sides are mutually perpendicular. Hence

$$\frac{A_2P}{C_4D_4} = \frac{A_2B_{10}}{C_4B_4} = \frac{B_{10}P}{B_4D_4} \,. \tag{7.150}$$

From (7.147) and (7.150), we obtain[19]

$$A_{10}B_{10} = A_{10}P + PB_{10} = \frac{A_2A_{10} \times A_6C_6}{A_6D_6} + \frac{B_4D_4 \times A_2B_{10}}{C_4B_4}$$

[19]It may be noted that equation (7.151) is essentially the *jīve-paraspara-nyāya* for the Rsines (which are half the full-chords) of the sum of 4 and 6 arc-bits, and the above constitutes one more demonstration of this rule.

$$= \left(\frac{1}{d}\right)(A_2A_{10} \times A_6C_6 + B_4D_4 \times A_2B_{10})\,. \quad (7.151)$$

Thus have been derived both the altitude and the base of the triangle $A_2A_{10}B_{10}$.

7.15 Area of a cyclic quadrilateral

The earlier result (7.141) gave the area of a cyclic quadrilateral in terms of the three diagonals and the circum-diameter. Though the three diagonals can be expressed in terms of the sides by equations (7.138) to (7.140), the expression for the area still involves the circum diameter. Now is shown a way of calculating the area of a cyclic quadrilateral given only the four sides (without employing the diameter), so that this expression for the area can also be used in conjunction with (7.138)–(7.141), to obtain the circum-diameter also in terms of the sides.

The final result to be derived is given by the rule *sarvadoryutidalam...*(*Līlāvatī*, 169), which may be expressed as follows using the same notation used in equations (7.138)–(7.140):[20]

$$A = [(s-a)(s-b)(s-c)(s-d)]^{\frac{1}{2}}, \quad (7.152)$$

where, the semi-perimeter s is given by

$$s = \frac{(a+b+c+d)}{2}\,. \quad (7.153)$$

A problem given in *Līlāvatī* 171, is also cited in this connection, where $a = 75,\ b = 40,\ c = 51$ and $d = 68$. The corresponding first diagonal (which may be calculated by (7.138)) is noted to be 77. This example is

[20]Combining (7.152) with (7.138) - (7.141),we are lead to the following formula for the circum-radius of a cyclic quadrilateral, in terms of its sides (cited by *Parameśvara* in his commentary *Vivaraṇa* on *Līlāvatī*; also cited by *Śaṅkara* in his commentary *Kriyākramakarī* on *Līlāvatī*):

$$r = \left(\frac{1}{4}\right)\left[\frac{(ab+cd)(ac+bd)(ad+bc)}{(s-a)(s-b)(s-c)(s-d)}\right]^{\frac{1}{2}}.$$

cited to illustrate the set up or the kind of arrangement of the sides that is being considered here.[21]

7.15.1 Area in terms of the *Lamba-nipātāntara* and *Lamba-yoga* (interstice between the altitudes and the sum of altitudes)

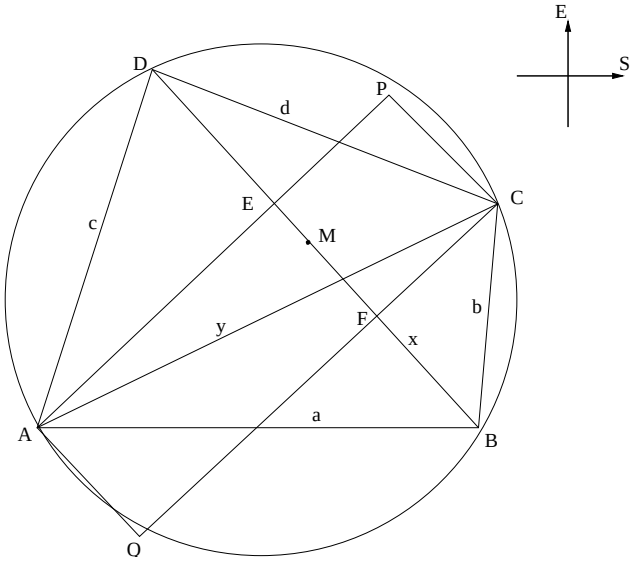


Figure 7.15: Area of a cyclic quadrilateral.

In Figure 7.15 we consider a cyclic quadrilateal $ABCD$ of the type mentioned above. Draw the altitudes AE and CF to the diagonal DB, which may be considered as the base of the two triangles CDB and ADB. Produce these altitudes such that they form the opposite sides of a rectangle, whose length $AP = CQ$ is the sum of the altitudes (*lamba-yoga*) and breadth $CP = AQ$ is equal to EF, the interstice (*lamba-nipātāntara*) between the feet of the perpendiculars AE and CF.

The procedure for deriving the area of the cyclic quadrilateral is the same as followed earlier, namely express it as the sum of the areas of the two triangles into which it is divided by a diagonal and calculate the area of the

[21] Note that this arrangement differs slightly from the arrangement considered in Section 7.10, where it was assumed that $a > b > d > c$. Now we have $a > d > c > b$.

triangles by using the diagonal as the base and the perpendiculars drawn to it as the altitudes. Only difference, now, is that the sum of these altitudes (*lamba-yoga*) will be obtained in terms of the sides of the quadrilateral.

The square of the area of the quadrilateral is given by

$$\begin{aligned} A^2 &= \left[\frac{1}{2}BD \times (AE + CF)\right]^2 \\ &= \left(\frac{1}{4}\right) BD^2 \times AP^2. \end{aligned} \tag{7.154}$$

Now the *lamba-yoga* AP and the *lamba-nipātāntara* EF are related to the other diagonal AC as they are the sides of the rectangle of which AC is the diagonal. Hence

$$AP^2 + EF^2 = AC^2. \tag{7.155}$$

Hence the area of the cyclic quadrilateral can be expressed in terms of the diagonals BD, AC and the *lamba-nipātāntara* EF as follows:

$$A^2 = \left(\frac{1}{4}\right) [BD^2 \times (AC^2 - EF^2)]. \tag{7.156}$$

7.15.2 Derivation of the *Lamba-nipātāntara*

In Figure 7.15 we have the situation (same as in the example of *Līlāvatī*, cited above) that $AB > AD$ and $CD > CB$. That is, in the triangles ABD and CBD, the longer sides AB and CD meet different vertices of the base BD . Then the feet of the perpendiculars AE and CF fall on different sides of the mid-point M of the diagonal BD. Then the *lamba-nipātāntara*, EF, is given by the sum of the distances of the feet of the perpendiculars from the mid-point M.

$$EF = ME + MF. \tag{7.157}$$

The distance of the foot of the perpendicular to the centre of the base is nothing but half the difference between the base-segments (*ābādhā*) of a triangle. That is

$$ME = \frac{1}{2}(BE - DE). \tag{7.158}$$

$$MF = \frac{1}{2}(DF - BF). \tag{7.159}$$

The difference between the base-segments in a triangle is nothing but the difference of their squares divided by the base, which is the sum of the base-segments. And as has been noted in equation (7.109), the difference in the squares of the base-segments is equal to the difference between the squares of the sides of the triangle. Thus

$$ME = \frac{1}{2}(BE - DE) = \frac{1}{2}\frac{(AB^2 - DA^2)}{BD} , \tag{7.160}$$

$$MF = \frac{1}{2}(DF - BF) = \frac{1}{2}\frac{(CD^2 - BC^2)}{BD} . \tag{7.161}$$

Hence, we obtain from (7.157)–(7.161) that the *lamba-nipātāntara* is given by

$$EF = \frac{1}{2}\frac{(AB^2 + CD^2) - (BC^2 + DA^2)}{BD} . \tag{7.162}$$

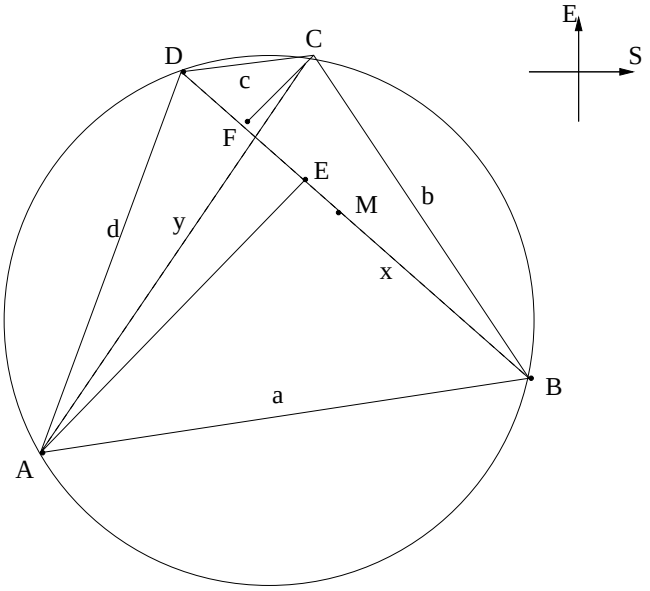


Figure 7.16: Area of a cyclic quadrilateral.

If we consider a different situation as shown in Figure 7.16, where though $AB > AD$ as before, we have $BC > CD$. That is, in the triangles ABD and CBD, the longer sides AB and BC meet at the same vertex B of the base BD. Then the feet of the perpendiculars AE and CF fall on the same side of the mid-point M of the diagonal BD. Then the *lamba-nipātāntara*, EF,

is given by the difference of the distances of the feet of the perpendiculars from the mid-point M.

$$EF = MF - ME. \tag{7.163}$$

Again as before we get

$$ME = \frac{1}{2}(BE - DE) = \frac{1}{2}\frac{(AB^2 - DA^2)}{BD}\,, \tag{7.164}$$

$$MF = \frac{1}{2}(BF - DF) = \frac{1}{2}\frac{(BC^2 - CD^2)}{BD}\,. \tag{7.165}$$

Thus, from (7.163)–(7.165), we again obtain

$$EF = \frac{1}{2}\ \frac{(AB^2 + CD^2) - (BC^2 + DA^2)}{BD}\,. \tag{7.166}$$

This is the same as (7.162). Thus we see that in each case the *lamba-nipātāntara* is given by the half the difference between the sums of the squares of the opposite sides of the cyclic quadrilateral divided by the diagonal.

7.15.3 First result for the area

From the above results (7.156) and (7.166), we obtain the following for the square of the area of the cyclic quadrilateral:

$$\begin{aligned} A^2 &= \left(\frac{1}{4}\right)[BD^2(AC^2 - EF^2)] \\ &= \left(\frac{1}{4}\right)\left[BD^2 \times AC^2 - \frac{1}{4}\{(AB^2 + CD^2) - (BC^2 + DA^2)\}^2\right] \\ &= \left[BD \times \frac{AC}{2}\right]^2 \\ &\quad - \left[\left\{\left(\frac{AB}{2}\right)^2 + \left(\frac{CD}{2}\right)^2\right\} - \left\{\left(\frac{BC}{2}\right)^2 + \left(\frac{DA}{2}\right)^2\right\}\right]^2 \end{aligned} \tag{7.167}$$

Thus in order to obtain the square of the area of a cyclic quadrilateral half the product of the diagonals and the difference of the sums of the squares of half the opposite sides can be squared and one subtracted from the other. This is as per the rule given by the verse *pratibhujadala...*

7.15.4 Second result for the area

Once again, adopting the notation $AB = a,\ BC = b,\ CD = c$ and $DA = d$ (not to be confused with the diameter) for the sides and $BD = x,\ AC = y$ for the diagonals, equation (7.167) for the area becomes

$$A^2 = \left[\frac{xy}{2}\right]^2 - \left[\left\{\left(\frac{a}{2}\right)^2 + \left(\frac{c}{2}\right)^2\right\} - \left\{\left(\frac{b}{2}\right)^2 + \left(\frac{d}{2}\right)^2\right\}\right]^2 . \tag{7.168}$$

Now, using (7.138) and (7.139), (7.168) becomes

$$A^2 = \left[\frac{(ac+bd)}{2}\right]^2 - \left[\left\{\left(\frac{a}{2}\right)^2 + \left(\frac{c}{2}\right)^2\right\} - \left\{\left(\frac{b}{2}\right)^2 + \left(\frac{d}{2}\right)^2\right\}\right]^2 . \tag{7.169}$$

We can express the above difference of squares in the form of a product of sum and difference and obtain

$$\begin{aligned} A^2 &= \left[\frac{(ac+bd)}{2} + \left\{\left(\frac{a}{2}\right)^2 + \left(\frac{c}{2}\right)^2\right\} - \left\{\left(\frac{b}{2}\right)^2 + \left(\frac{d}{2}\right)^2\right\}\right] \times \\ &\quad \left[\frac{(ac+bd)}{2} - \left\{\left(\frac{a}{2}\right)^2 + \left(\frac{c}{2}\right)^2\right\} + \left\{\left(\frac{b}{2}\right)^2 + \left(\frac{d}{2}\right)^2\right\}\right] \\ &= \left[\left\{\left(\frac{a}{2}\right) + \left(\frac{c}{2}\right)\right\}^2 - \left\{\left(\frac{b}{2}\right) - \left(\frac{d}{2}\right)\right\}^2\right] \times \\ &\quad \left[\left\{\left(\frac{b}{2}\right) + \left(\frac{d}{2}\right)\right\}^2 - \left\{\left(\frac{a}{2}\right) - \left(\frac{c}{2}\right)\right\}^2\right] . \end{aligned} \tag{7.170}$$

7.15.5 Final result for the area

Again expressing each of the factors in (7.170), which is a difference of squares, as a product of a sum and a difference we obtain

$$\begin{aligned} A^2 &= \left(\frac{a}{2} + \frac{c}{2} + \frac{b}{2} - \frac{d}{2}\right)\left(\frac{a}{2} + \frac{c}{2} + \frac{d}{2} - \frac{b}{2}\right) \times \\ &\quad \left(\frac{b}{2} + \frac{d}{2} + \frac{a}{2} - \frac{c}{2}\right)\left(\frac{b}{2} + \frac{d}{2} + \frac{c}{2} - \frac{a}{2}\right) . \end{aligned} \tag{7.171}$$

If we denote the semi-perimeter of the cyclic quadrilateral by

$$s = \frac{a}{2} + \frac{b}{2} + \frac{c}{2} + \frac{d}{2} , \tag{7.172}$$

then the above expression (7.171) for the area reduces to

$$A = [(s-a)(s-b)(s-c)(s-d)]^{\frac{1}{2}}. \tag{7.173}$$

This is the result given by the rule *samadoryutidalam...* (*Līlāvatī*, 169).

7.15.6 Area of triangles

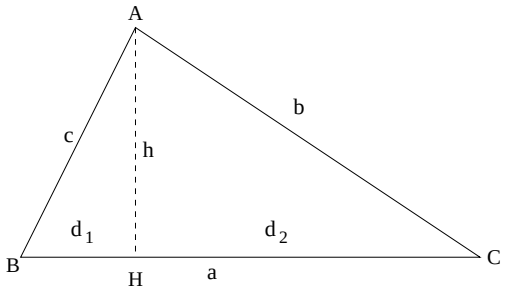


Figure 7.17: Area of a triangle.

In the case of triangle also, the area can be expressed in a manner similar to (7.173). In Figure 7.17, ABC is a triangle, AH is the altitude from vertex A to the base BC. We adopt the notation $BC = a$, $AC = b$ and $AB = c$ for the sides, $AH = h$ for the altitude and $BH = d_1$ and $CH = d_2$, for the base-segments (*ābādhā*). Let the semi-perimeter s be given by

$$s = \frac{(a+b+c)}{2}\ . \tag{7.174}$$

Now it will be shown that square of the area of the triangle is given by

$$A^2 = [s(s-a)(s-b)(s-c)]. \tag{7.175}$$

It is noted that

$$\begin{aligned} s(s-a) &= \left[\frac{(b+c)}{2}\right]^2 - \left(\frac{a}{2}\right)^2 \\ &= \left[\frac{(b+c)}{2}\right]^2 - \left[\frac{(d_1+d_2)}{2}\right]^2, \end{aligned} \tag{7.176}$$

may be considered to be nearly equal to the square of the perpendicular, which is given by

$$h^2 = b^2 - d_2^2$$

$$\begin{aligned} &= c^2 - d_1^2 \\ &= \frac{(b^2 + c^2)}{2} - \frac{(d_1^2 + d_2^2)}{2} \,. \end{aligned} \tag{7.177}$$

Similarly, it is also noted that

$$(s-b)(s-c) = \left(\frac{a}{2}\right)^2 - \left[\left(\frac{b}{2}\right) - \left(\frac{c}{2}\right)\right]^2, \tag{7.178}$$

may be considered to be nearly equal to the square of half the base. It is therefore argued that the expression (7.175) for the square of the area may be considered to be nearly the same as the one which was obtained earlier in Section 7.9:

$$A^2 = h^2 \left(\frac{a}{2}\right)^2 . \tag{7.179}$$

The exact equivalence of the expressions (7.175) and (7.179) is now proved as follows: From (7.176) and (7.177) we note that

$$s(s-a) = h^2 + \left[\frac{(d_2 - d_1)}{2}\right]^2 - \left[\frac{(b-c)}{2}^2\right]^2 . \tag{7.180}$$

Thus the product of the first two factors in the right hand side of (7.175) is in excess of the square of the altitude, the first factor in the right hand side of equation (7.179), by the amount given in equation (7.180). Equation (7.178) itself gives the amount by which the product of last two factors in the right hand side of (7.175) is deficient from the square of half the base, the second factor on the right hand side of (7.179.

Now, we recall that the difference in the squares of the base-segments ($\bar{a}b\bar{a}dh\bar{a}$) is also the difference in the squares of the sides

$$d_2^2 - d_1^2 = b^2 - c^2. \tag{7.181}$$

Therefore

$$\frac{(d_2 - d_1)}{(b+c)} = \frac{(b-c)}{(d_2 + d_1)} \,. \tag{7.182}$$

The above equation can be viewed as a *trairāśika*, rule of proportion, which will therefore hold for half their squares and even their sums and differences. Thus from (7.182), we obtain

$$\frac{\left[\frac{(d_2 - d_1)}{2}\right]^2}{\left[\frac{(b+c)}{2}\right]^2} = \frac{\left[\frac{(b-c)}{2}\right]^2}{\left[\frac{(d_2 + d_1)}{2}\right]^2} \,. \tag{7.183}$$

From (7.183) it follows that

$$\frac{\left[\frac{(b-c)}{2}\right]^2}{\left[\frac{(d_2+d_1)}{2}\right]^2} = \frac{\left[\left(\frac{(d_2-d_1)}{2}\right)^2 - \left(\frac{(b-c)}{2}\right)^2\right]}{\left[\left(\frac{(b+c)}{2}\right)^2 - \left(\frac{(d_2+d_1)}{2}\right)^2\right]}\,, \tag{7.184}$$

or, equivalently

$$\frac{\left[\frac{(b-c)}{2}\right]^2}{\left(\frac{a}{2}\right)^2} = \frac{\left[\left(\frac{(d_2-d_1)}{2}\right)^2 - \left(\frac{(b-c)}{2}\right)^2\right]}{\left[\left(\frac{(b+c)}{2}\right)^2 - \left(\frac{(d_2+d_1)}{2}\right)^2\right]}\,. \tag{7.185}$$

Now, in order to show that (7.175) follows from (7.179), let us first consider an example. Suppose we want to multiply 5 by 3. But if instead of 5, say $6 = 5 + 5.\frac{1}{5}$, is chosen as the multiplicand, then the multiplier 3 will have to be reduced by an amount $\frac{3}{6}$, obtained by dividing it by a divisor one larger than the divisor used in the additive term to the multiplicand.[22] That is

$$5 \times 3 = 15 = \left[5 + 5\left(\frac{1}{5}\right)\right]\left[3 - 3\left(\frac{1}{(5+1)}\right)\right] = 6 \times 2\frac{1}{2}\,. \tag{7.186}$$

Now, we start with (7.179), and show how it leads to (7.175), by employing the method of equation (7.186). It follows from (7.186) that

$$\begin{aligned} A^2 &= h^2\left(\frac{a}{2}\right)^2 \\ &= \left[h^2 + \left(\frac{(d_2-d_1)}{2}\right)^2 - \left(\frac{(b-c)}{2}\right)^2\right] \times \\ &\quad \left[\left(\frac{a}{2}\right)^2 - \left(\frac{a}{2}\right)^2 \frac{\left(\frac{(d_2-d_1)}{2}\right)^2 - \left(\frac{(b-c)}{2}\right)^2}{\left(\frac{(d_2-d_1)}{2}\right)^2 - \left(\frac{(b-c)}{2}\right)^2 + h^2}\right]. \end{aligned} \tag{7.187}$$

[22] This has been explained in Section 1.6 as the third special method of multiplication. Equation (7.187) follows directly from the identity

$$pq = (p+r)\left(q - \frac{qr}{p+r}\right).$$

Now, by employing (7.176) and (7.180), (7.187) becomes

$$A^2 = s(s-a)\left[\left(\frac{a}{2}\right)^2 - \left(\frac{a}{2}\right)^2 \frac{\left(\frac{(d_2-d_1)}{2}\right)^2 - \left(\frac{(b-c)}{2}\right)^2}{\left(\frac{(b+c)}{2}\right)^2 - \left(\frac{(d_1+d_2)}{2}\right)^2}\right]. \qquad (7.188)$$

If we now employ (7.185) in (7.188), we get

$$\begin{aligned} A^2 &= s(s-a)\left[\left(\frac{a}{2}\right)^2 - \left(\frac{a}{2}\right)^2 \left(\frac{\left(\frac{(b-c)}{2}\right)^2}{\left(\frac{a}{2}\right)^2}\right)\right] \\ &= s(s-a)\left[\left(\frac{a}{2}\right)^2 - \left(\frac{(b-c)}{2}\right)^2\right]. \end{aligned} \qquad (7.189)$$

Now by using (7.178) in (7.189), we finally obtain the desired relation (7.175) for the area of the triangles as given by the rule *sarvadoryutidalam...* (*Līlāvatī* 169).

$$A^2 = [s(s-a)(s-b)(s-c)]. \qquad (7.190)$$

7.16 Derivation of the *Sampāta-śara*

In Figure 7.18 we consider two circles intersecting each other. Let AB and CD be the overlapping diameters and EF the common chord which is bisected perpendicularly by the diameters. CB, the portion of the diameters between the two circles is called the *grāsa*, the erosion and GB and GC are the the *sampāta-śara*, arrows of intercepted arc. The smaller circle will have the larger intercepted arc and vice versa.

If we employ the rule *vyāsāt śaronāt...*(*Līlāvatī*, 204), that the product of the arrow and diameter minus arrow is the square of the perpendicularly bisected chord, we have

$$AG \times GB = EG^2 = CG \times GD\ . \qquad (7.191)$$

Therefore, we have

$$(AB - GB) \times GB = (CB - GB)(CD - CB + GB). \qquad (7.192)$$

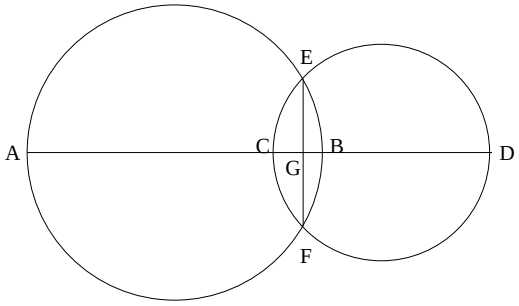


Figure 7.18: *Sampāta-śara.*

Therefore

$$AB \times GB = CB \times (CD - CB) + CB \times GB - (CD - CB)GB \ . \quad (7.193)$$

Thus

$$GB = \frac{(CD - CB) \times CB}{[(AB - CB) + (CD - CB)]} \ . \quad (7.194)$$

Similarly the other arrow of the intercept is given by

$$GC = \frac{(AB - CB) \times CB}{[(AB - CB) + (CD - CB)]} \ . \quad (7.195)$$

Equations (7.194) and (7.195) are according to the rule *śarone dve vṛtte...* (*Āryabhaṭīya, Gaṇita*, 18).

7.17 Derivation of the shadow

In Figure 7.19 are shown the shadows cast by a lamp LM when a gnomon of height 12 units is placed in two different positions AH and PQ. Let HB, QR be the corresponding shadows and AB, PR, the corresponding shadow hypotenuses (*chāyā-karṇa*). Now, mark C on the ground such that $HC = QR$. Then $AC = PR$.

The problem is to calculate the shadows given the difference between them as also the difference between the shadow-hypotenuses. We can see in Figure 7.19 that in the triangle ABC, the sides AB, AC are the shadow-hypotenuses, the altitude $AH = 12$ is the gnomon and the base-segments

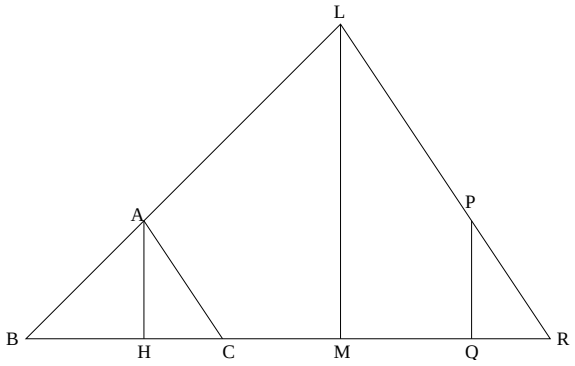


Figure 7.19: Derivation of the shadow.

BH and HC are the shadows. Thus the problem may be considered to be one of determining the base-segments given their difference, as also the altitude and the difference of sides in the triangle ABC. We shall adopt the notation of Section 7.15.6, namely $AB = c,\ AC = b,\ BC = a,\ AH = h,\ BH = d_1,\ CH = d_2$.

As before, we start with the relation that the difference in the squares of the sides is equal to the difference in the squares of the base segments.

$$d_1^2 - d_2^2 = c^2 - b^2. \tag{7.196}$$

which is equivalent to the relation

$$\frac{(d_1 + d_2)}{(c - b)} = \frac{(c + b)}{(d_1 - d_2)}\ . \tag{7.197}$$

Viewing the above relation as a *trairāśika*, rule of proportion, we can derive the relation

$$\begin{aligned} \frac{(d_1 + d_2)^2}{(c - b)^2} &= \frac{(c + b)^2}{(d_1 - d_2)^2} \\ &= \frac{[(c + b)^2 - (d_1 + d_2)^2]}{[(d_1 - d_2)^2 - (c - b)^2]}\ . \end{aligned} \tag{7.198}$$

If we now recall that the altitude is given by the relation (7.177)

$$h^2 = \frac{(b^2 + c^2)}{2} - \frac{(d_1^2 + d_2^2)}{2}\ , \tag{7.199}$$

it follows that

$$\frac{(d_1+d_2)^2}{(c-b)^2} = \frac{[4h^2+(d_1-d_2)^2-(c-b)^2]}{[(d_1-d_2)^2-(c-b)^2]}\ . \tag{7.200}$$

Since the height of the gnomon $h = 12$, (7.198) becomes

$$d_1+d_2 = (c-b)\left[1+\frac{576}{[(d_1-d_2)^2-(c-b)^2]}\right]^{\frac{1}{2}}. \tag{7.201}$$

Equation (7.201) gives the sum of the shadows d_1+d_2 given the difference d_1-d_2 between the shadows as also the difference $c-b$ between the shadow-hypotenuses. Then the shadows d_1 and d_2 can be obtained and this is as per the rule *chāyayoḥ karṇayor*...(*Līlāvatī*, 238).

7.18 Surface area of a sphere

The area of a sphere will now be obtained using the earlier result (7.50) for the *piṇḍajyā*-s, Rsines, in terms of *khaṇḍa-jyāntara*-s, second order Rsine-differences, and the fact that knowing the circumference and diameter of a circle the circumference or the diameter of any desired circle can be obtained in terms of the other by the rule of three.

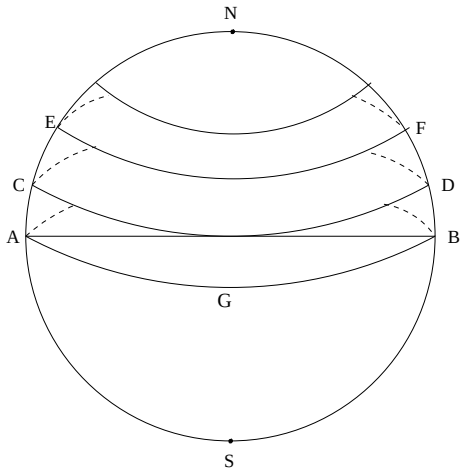


Figure 7.20: Surface area of a sphere.

In Figure 7.20 is shown a sphere with the north-south circle $NASBN$ perpendicular to the equatorial plane AGB. Divide the north-south circle into a large number of equal arc-bits and divide the sphere into slices of equal thickness by planes parallel to the equatorial plane passing through the junctions of the arc-bits. If we cut the surface of any slice and stretch the piece flat, what results is a trapezium, as shown in Figure 7.21. If we cut the triangle at one edge and fix it at the other suitably, we get a rectangle whose width is the thickness of the slices and length is the mean of the perimeters of the top and bottom circles of the slice. The surface area of the sphere is the sum of the areas of these rectangles.

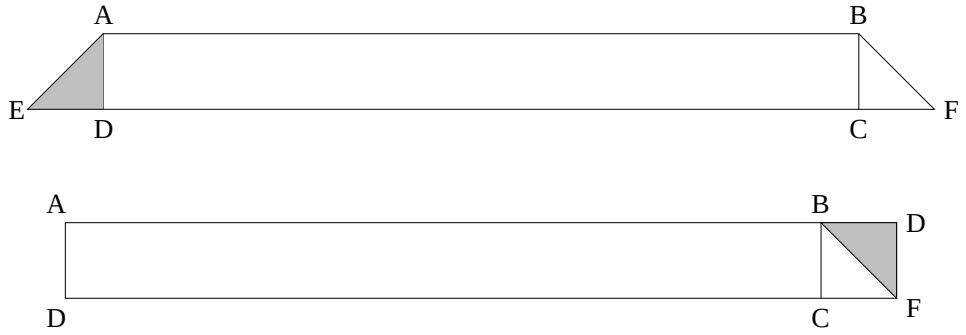


Figure 7.21: Surface area of the slices.

Let r be the radius of the sphere and C the circumference of any a great circle. The radius of the middle of the j-th slice will be a half chord at the middle of the corresponding arc-bit in the north-south circle. When the number of slices are very large we can approximate the half-chords at the middle of the arc-bits by the corresponding *piṇḍajyā*-s B_j at the junction of the arc-bits. The common thickness of the slices, which we shall denote by Δ, is nothing but the Rsine of the arc-bit.

Now, the area of the j-th rectangle will be $\left(\frac{C}{r}\right)\ B_j\ \Delta$. Therefore the surface area of the northern hemisphere is given by

$$\frac{S}{2} = \left(\frac{C}{r}\right)(B_1 + B_2 + \ldots B_n)\Delta\ . \tag{7.202}$$

If we recall the relation (7.50) between the *piṇḍajyā*-s and the second order sine-differences, then the sum of Rsines in (7.202) can be expressed as a sum

of the second-order sine-differences

$$\begin{aligned} B_1 + B_2 + \ldots + B_n &= \left(\frac{r}{\alpha}\right)^2 [(\Delta_1 - \Delta_2) + (\Delta_2 - \Delta_3) + \\ \ldots + (\Delta_{n-1} - \Delta_n)] & \\ &= \left(\frac{r}{\alpha}\right)^2 (\Delta_1 - \Delta_n), \end{aligned} \tag{7.203}$$

where $\Delta_1,\ \Delta_2 \ldots$ are the Rsine-differences and α is the full-chord of the arc-bit. When n is very large, Δ_n becomes negligible[23] and the first Rsine, $B_1 = \Delta_1 = \Delta$, can be approximated by the arc-bit or by its full-chord α. Thus, when n is very large, (7.202) becomes

$$\begin{aligned} \frac{S}{2} &= \left(\frac{C}{r}\right)\left(\frac{r}{\alpha}\right)^2 (\Delta_1 - \Delta_n)\Delta \\ &\approx \left(\frac{C}{r}\right)\left(\frac{r}{\alpha}\right)^2 (\alpha)^2 = Cr\ . \end{aligned} \tag{7.204}$$

Thus the surface area of the sphere is given by the product of the circumference C and the diameter d

$$S = C \times d\ . \tag{7.205}$$

7.19 Volume of a sphere

To find the volume of sphere, it is again divided into large number of slices of equal thickness. The volume of each slice is the average of the areas of the top and bottom circles multiplied by the thickness. Thus we need to know the area of a circle which is derived first.

7.19.1 Area of a circle

As shown in Figure 7.22, cut each half of the circle into a large number sections from the centre to the circumference. Then as in Figure 7.23, spread

[23] This can be seen for instance from relation (7.49), which shows that Δ_n is proportional to the *piṇḍa-koṭi-jyā*, $K_{n-\frac{1}{2}}$ at the mid-point of the last arc-bit, and by noting that the Rcosine becomes negligible as n becomes very large.

these out and place the sections of the upper half so as to fit the interstices when the sections of the lower half are spread out. The area of the circle is equal to that of the resulting rectangle, namely half the circumference multiplied by the radius.

$$A = \frac{1}{2} C \times r \ . \tag{7.206}$$

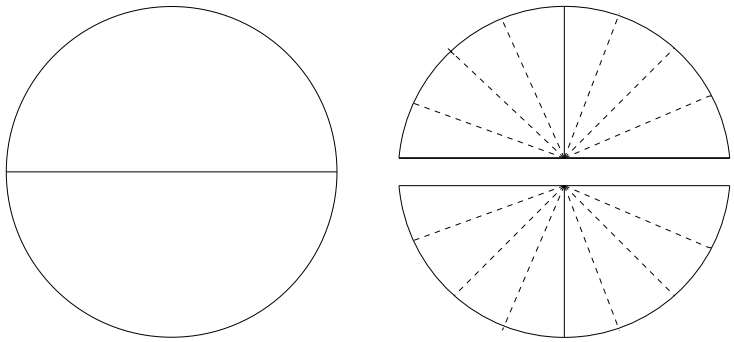


Figure 7.22: Division of a circle into sections.

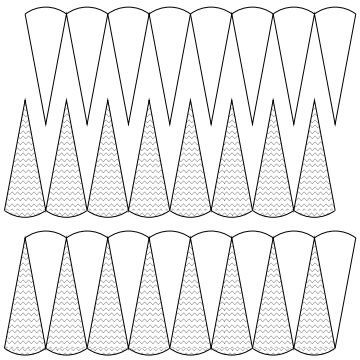


Figure 7.23: Area of a circle.

7.19.2 Derivation of the volume of a sphere

As before, let r be the radius of the sphere and C the circumference of a great circle. Again the half-chord B_j is the radius of the j-th slice into which the sphere has been divided. The corresponding circumference is $\left(\frac{C}{r}\right) B_j$ and from (7.206) the area of a circular section of this slice can be taken to be $\frac{1}{2}\left(\frac{C}{r}\right) B_j^2$. Therefore, if Δ is the thickness of the slices, then the volume of

the j-th slice is given by $\frac{1}{2}\left(\frac{C}{r}\right) B_j^2 \Delta$. To obtain the volume of the sphere, we need to therefore evaluate the sum of the squares of the Rsines B_j^2.

$$V \approx \frac{1}{2}\left(\frac{C}{r}\right)[B_1^2 + B_2^2 + \ldots B_n^2]\Delta \; . \tag{7.207}$$

In Figure 7.24, $AP = PB = B_j$ is the jth half-chord, starting from N, the north point. By the rule *vṛtte śarasaṃvargo...* (*Āryabhaṭīya, Gaṇita* 17),

$$\begin{aligned} B_j^2 &= AP \times PB = NP \times SP \\ &= \frac{1}{2}\left[(NP + SP)^2 - (NP^2 + SP^2)\right] \\ &= \frac{1}{2}\left[(2R)^2 - (NP^2 + SP^2)\right]. \end{aligned} \tag{7.208}$$

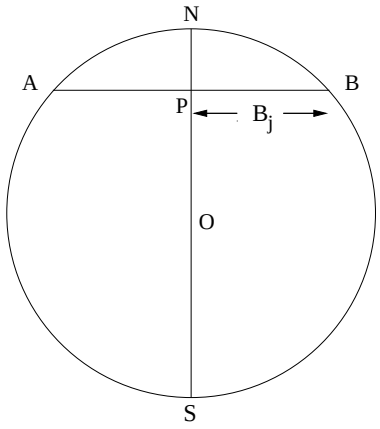


Figure 7.24: Half-chord square equals product of *śara*-s.

If $\Delta = 2\frac{r}{n}$, be the thickness of each slice, the j-th Rversine $NP = j\Delta$ and its complement $SP = (n - j)\Delta$. Hence, while summing the squares of the Rsines B_j^2, both NP^2 and SP^2 add to the same result. Thus, using (7.208), we can re-express (7.207) in the form

$$\begin{aligned} V \;\approx\; & \frac{1}{2}\left(\frac{C}{r}\right)\left(\frac{2r}{n}\right)\left(\frac{1}{2}\right)[(2r)^2 + (2r)^2 + \ldots + (2r)^2] \\ & -\frac{1}{2}\left(\frac{C}{r}\right)\left(\frac{2r}{n}\right)\left(\frac{1}{2}\right)\left(\frac{2r}{n}\right)^2 (2)[1^2 + 2^2 + ...n^2]. \end{aligned} \tag{7.209}$$

Recalling the result that, for large n, the sum of the squares (*varga-saṅkalita*)

is essentially one-third the cube of the number of terms, (7.209) becomes

$$\begin{aligned} V &= \left(\frac{C}{2r}\right)\left(4r^3 - \frac{8r^3}{3}\right) \\ &= \left(\frac{C}{6}\right)d^2. \end{aligned} \tag{7.210}$$

Thus has been shown that the volume of a sphere is one-sixth the circumference times the square of the diameter.